Products of Cosines: How the Entangling Layers of Quantum Classifier Heads Shape the CNN Representations Trained Through Them

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Abstract

When a convolutional network is trained end to end through a variational quantum circuit (VQC) classifier head, the head shapes the features the network learns, and we measure this with one common linear probe on the frozen bottleneck of each tuned network. On CIFAR-10 with 4,000 training images, an 8-qubit VQC head with CNOT rings lowers probe accuracy by 4.7 points relative to a linear head (5 seeds, every seed lower; a two-hidden-layer MLP probe finds the same gap), and removing the rings recovers 3.8 of those points. An exact identity organizes the pattern: with every variational rotation at zero, each readout is a product of cosines of the encoded inputs over a set fixed by the CNOT layers. Probe accuracy falls as the mean order of these products rises, from 73.6 at order 1 to 69.8 at the default order 2.5 and 60.1 at order 5.25; classical heads trained on the same products follow the same trend, the circuit ordering holds at twice the training budget and on MNIST, six new CNOT layouts show a strong inverse association with order (two rank reversals), as do six layouts under a second encoding, and a circuit that retains entangling gates but has product order 1 recovers 3.1 points. On an ImageNet-pretrained ResNet-18 the same ordering holds: at an approximately matched number of optimizer updates the default head costs 6–8 points with 1,000 to 10,000 training images, while the ring-free head stays within 1.4 points of a linear head; the ordering also holds on SVHN, with data re-uploading and on a pretrained MobileNetV3. Entangling gates alone do not set the size of the cost here: an entangling CZ ring of order 1 probes 0.4 points below the ring-free head, and multi-qubit readouts of a circuit without entangling gates fall on the order trend. We prove that, for independent inputs, high-order products that keep their magnitude drive the mean mass of their inputs in the graded middle of the encoding toward zero as order grows, and a preregistered intervention that lets saturated coordinates escape, with the forward pass unchanged, recovers half of the highest-order cost. Without entangling gates and with single-qubit readouts the head is exactly a classical per-coordinate sinusoidal model, and under matched per-head tuning a linear head has the highest mean test accuracy among linear, small-MLP and default VQC heads in all six matched dataset–size settings.

1. Introduction

A common near-term design for quantum image classification feeds a convolutional network into a small variational quantum circuit (VQC) that acts as the classifier head (Mari et al., 2020), building on variational quantum classifiers (Mitarai et al., 2018; Havlíček et al., 2019; Benedetti et al., 2019; Cerezo et al., 2021a). When the network is trained jointly *through* the circuit, the head does more than classify: its gradients shape the features the backbone learns. A comparison between a quantum head and a classical one therefore compares two things at once, the heads and the representations they induce. The question reaches beyond quantum machine learning: a head that distorts the features trained through it is a known failure mode of fine-tuning (Kumar et al., 2022), and here the distortion turns out to track a structural quantity that can be computed before training. Benchmarks of quantum classifiers typically score accuracy on fixed inputs (e.g. Fulton et al., 2026; Slattery et al., 2023); Bowles et al. (2024) also replace the quantum layer of a trained dressed hybrid network with a classical one and visualize how the learned layers transform the data. Controls that isolate a circuit inside a hybrid model score predictive performance (Kölle et al., 2024; Elhag et al., 2026). To our knowledge, none of these measures quantitatively what the quantum head does to the representation trained through it, or which part of the circuit is responsible.

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We answer both questions for a CNN with an 8-dimensional bottleneck and an 8-qubit head built from angle encoding $R_y(\pi \tanh z_j)$, four layers of trainable rotations each followed by a ring of CNOTs, and a linear readout of $\langle Z_k \rangle$. After joint training we freeze each backbone and fit the same linear probe to its bottleneck, so that probe accuracy compares the representations directly. The default VQC head lowers this probe accuracy, and the size of the drop follows the circuit's CNOT structure. An exact identity defines a structural descriptor whose order tracks this cost: when every trainable rotation is zero, which is near where training starts, each readout is $\langle Z_k \rangle = \prod_{j \in S_k} \cos(\pi \tanh z_j)$, a product of cosines over a set S_k of inputs fixed by the CNOT layers alone (Figure 1a). The mean size of these sets, the *product order*, is 1 without CNOTs, 2.5 for the default circuit and 5.25 when a single ring follows the last layer.

Our contributions are:

1. **A tuning audit** (Section 4). Under a revised protocol (longer training, validation-based selection, each head's learning rate tuned), the VQC's apparent advantage over a near-parameter-matched MLP on MNIST shrinks from up to 18 points to a largest observed mean of 3.3. On CIFAR-10 the VQC still trails the MLP by 3.1–9.2 points, and among the linear, MLP and default VQC heads the linear head has the highest mean test accuracy in all six dataset–size settings.

2. **A representation cost tied to the CNOT layers** (Sections 5.1 and 5.2). On CIFAR-10 with 4,000 training images the default VQC head lowers bottleneck probe accuracy by 4.7 ± 0.9 points relative to a linear head (5 seeds), and a two-layer MLP probe finds the same gap, so the loss is not just one of linear readability. Removing the CNOT rings recovers 3.8 ± 1.2 points. On a pretrained ResNet-18 the default head costs 6–8 points at 1,000 to 10,000 images at an approximately matched number of updates (22.3 at 1,000 with a fixed 30 epochs); the ring-free head stays within 1.4 points of linear (Section 5.6). Five controls leave a substantial gap (Section 5.5).

3. **A structural account** (Sections 5.3 and 5.4). Probe accuracy falls with product order across eleven heads (Figure 1b). Classical heads trained on a circuit's zero-rotation products, with no circuit, follow the same trend; on the small CNN the default circuit's products match the VQC (-1.0 ± 2.2), and on the pretrained backbone they reproduce the ordering. The ordering replicates on MNIST and on six new layouts (Spearman -0.89, two rank reversals), under predictions logged before but committed after those runs (Section U), and under committed predictions with a second encoding (-0.94) and at 12 qubits. A circuit that keeps its CNOTs but has product order 1 recovers 3.1 ± 1.2 points over the default. Entangling gates alone do not set the size of the cost here: a CZ ring, which entangles at order 1, probes 0.4 points below the ring-free head, and two- and three-body readouts of a circuit with no entangling gate land about 0.1 points from the order trend (Table 2). Without entangling gates and with single-qubit readouts the head is exactly a classical per-coordinate sinusoidal model, and classical heads of that form come within 0.3–0.9 probe points of it.

4. **An account** (Section 5.7). We prove that, for independent inputs, a high-order product readout that keeps its magnitude must drive the mean mass of its inputs in the graded middle of the encoding toward zero as order grows (a signal budget; on the trained bottlenecks the products keep their magnitude, but the bound is loose at these orders), and find that the CNNs' bottlenecks saturate early and stay saturated at high order. A pre-registered intervention that leaves the forward pass unchanged but lets saturated coordinates escape recovers 6.8 of the 13.5 last-ring points; the order cost that remains, and an unbounded encoding that keeps two thirds of the slope, are consistent with the budget, though we do not show that it mediates the cost.

2. Related Work

Benchmarking quantum classifiers and the role of entanglement. Large benchmarks find that simple classical models generally match or beat variational quantum classifiers, and that removing entanglement often does not hurt (Bowles et al., 2024); quantum fidelity kernels show a similar lack of advantage on classical data (Slattery et al., 2023). Closest to our setting, Fulton et al. (2026) show that classical $\mathrm{SU}(2)$ models match or exceed shallow VQCs on vision benchmarks and that ring entanglement can degrade accuracy on fixed features. Recent hybrid-network studies reach related conclusions with matched preprocessing and latent width (Freinberger & Moser, 2026) or matched width and parameter count (Park et al., 2026), and controlled benchmarks of quantum transfer learning find the outcome strongly configuration-dependent (Innan et al., 2026). Other work trains hybrid models end to end and studies the learned representation: Bowles et al. (2024) visualize the layers of a trained dressed hybrid network, Kashif & Shafique (2026) analyze expressibility and trainability under joint training, and quantum self-supervised learning evaluates frozen hybrid encoders with linear probes (Jaderberg et al., 2022; Edwards et al., 2026). We probe the supervised bottleneck *before* the circuit, the representation the circuit's gradients shape, and relate its accuracy to a quantity, product order, read off the CNOT pattern.

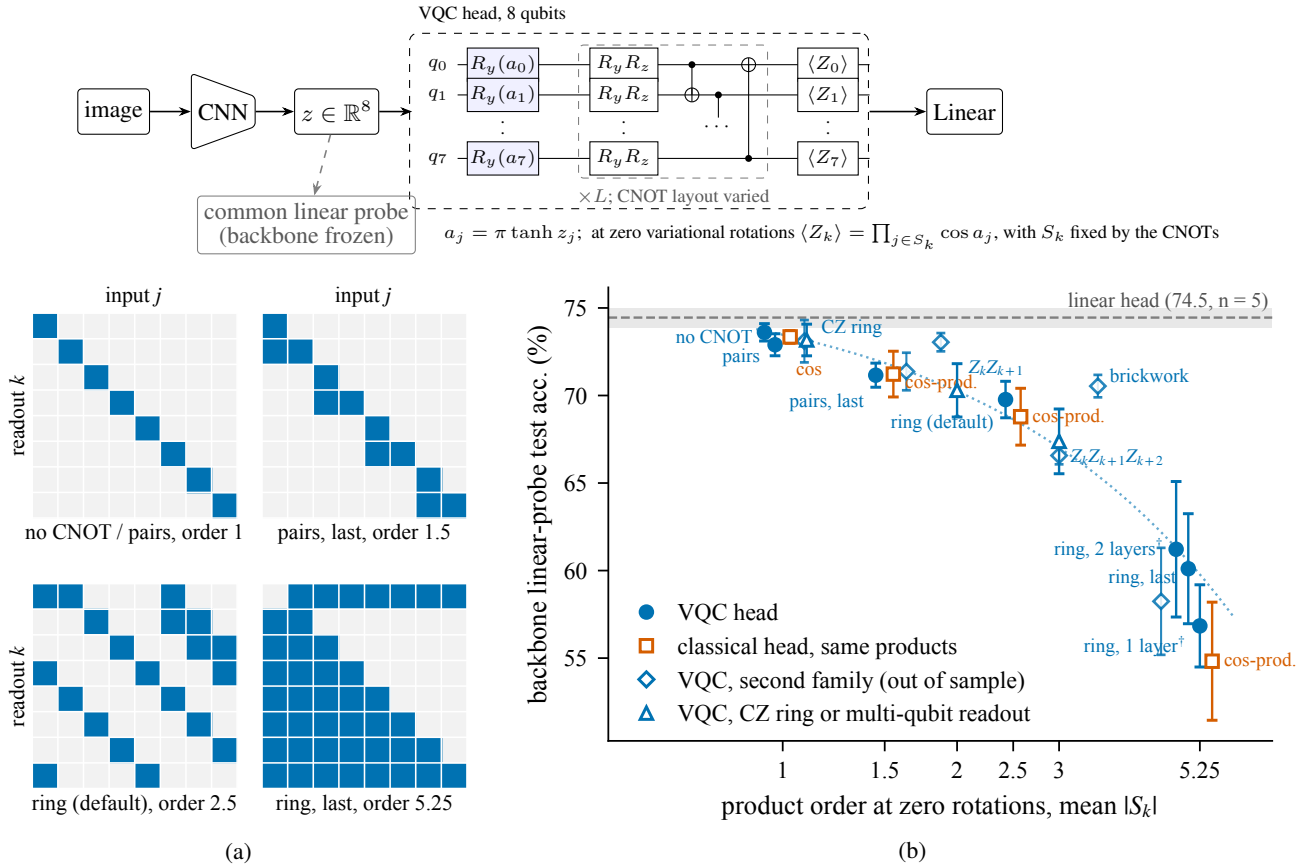


Figure 1. **(a)** A CNN feeds an 8-dimensional bottleneck z into an 8-qubit VQC head (default: four R_yR_z layers, each followed by a CNOT ring $(q, q{+}1 \bmod 8)$; variants place CNOTs after the last layer only ("last"), use $L \in \{1, 2\}$ layers, or replace the ring by disjoint pairs $(0,1),(2,3),(4,5),(6,7)$). With every variational rotation at zero, each readout is exactly $\langle Z_k \rangle = \prod_{j \in S_k} \cos(\pi \tanh z_j)$; the grids mark $j \in S_k$ in blue (rows: readouts k, columns: inputs j). Pairs after every layer cancel at zero rotations, so their supports equal the no-CNOT ones. Product order is $\frac{1}{8}\sum_k |S_k|$. The backbone is evaluated by one common linear probe on the frozen bottleneck. **(b)** CIFAR-10 with 4,000 training images: probe test accuracy against product order (log axis) for VQC heads (filled) and classical heads trained end to end on the zero-rotation products of a circuit (open; at order 5.25 these are the supports of "ring, last" and "ring, 1 layer"; "ring, 2 layers" has different supports of the same mean order). Mean ± SD over 5 seeds (†: 3); head learning rate selected on mean validation accuracy over seeds 0–2, seeds 3–4 run at that choice. Dashed line and band: linear head. Points at one order are offset horizontally. The plot shows an association; order co-varies with support overlap. Open diamonds: six CNOT layouts outside the ring family (5 seeds, Section G), whose probe accuracy was predicted before training (in our log, Section U) from the dotted fit over the ring-family (seeds 0–2) VQC circuits (linear in order, hence curved on the log axis); brickwork lies 5 points above its prediction. Open triangles: a CZ ring (entangling, order 1) and the CNOT-free circuit read out with Z_kZ_{k+1} or $Z_kZ_{k+1}Z_{k+2}$ (orders 2 and 3), pre-registered, 5 seeds (Table 2).

Isolating the circuit inside hybrid models. Kölle et al. (2024) separate classical and quantum contributions in hybrid architectures, building on sequential training that first fits a classical surrogate and then swaps in the circuit (Altmann et al., 2023). Elhag et al. (2026) delete, freeze or corrupt the circuit in quantum message passing and compare it with parameter-matched classical baselines and an exact classical surrogate for its bare R_y measurement, under a $\pi \tanh$ encoding. Encodings in hybrid convolutional models have also been compared (Monnet et al., 2024). These controls score predictive performance. We add a common probe on the jointly trained bottleneck and structural controls (Sections 5.3 and 5.5), none of which varies product order alone (the cancelling pairs also halve the CNOT count).

Fourier structure, classical surrogates and head-induced feature distortion. Angle-encoded circuits compute truncated Fourier series of their encoded angles (Schuld et al., 2021), including under data re-uploading (Pérez-Salinas et al., 2020), which classical models can represent and emulate (Schreiber et al., 2023); sinusoidal features also appear in classical networks as periodic activations and Fourier feature maps (Sitzmann et al., 2020; Tancik et al., 2020). Our zero-rotation identity is an elementary instance of this structure, used as an explanatory variable: it shows which

products the CNOT layers create, and their order tracks a measured cost. In classical deep learning, Kumar et al. (2022) show that the head used during fine-tuning distorts pretrained features, and linear probes are a standard tool for reading what an intermediate layer encodes (Alain & Bengio, 2017). We apply the probe to the quantum-head setting, where the head is a fixed-form circuit rather than a learned classical layer.

Vanishing gradients through products and saturating maps. In sufficiently random circuits gradients vanish exponentially with qubit count (McClean et al., 2018), and in shallow alternating layered circuits with locally random blocks global costs, such as tensor products of observables over many qubits, do so where local costs need not (Cerezo et al., 2021b). Our signal budget (Proposition 1) is related but constrains the network feeding the circuit. Our saturation test adapts the straight-through estimator (Bengio et al., 2013).

Position of this work. That entanglement often does not help, and that trigonometric surrogates can stand in for simple circuits, are known. New here are (i) a probe, shared across heads, of the bottleneck a quantum head induces under joint training, (ii) product order, which orders this cost across layouts, gate families and readouts, and (iii) a conditional account of it, tested by intervention.

3. Setup

Model. A CNN backbone maps an image to a bottleneck $\mathbf{z} \in \mathbb{R}^8$ (MNIST: three convolutional blocks, about 94K parameters; CIFAR-10: four blocks, about 0.96M; Section 5.6 also uses an ImageNet-pretrained ResNet-18). Each head is trained jointly with its own copy of the backbone. The *linear* head is a single dense layer (90 parameters); *MLP-8* has one hidden layer of width 8 with ReLU (162). The default *VQC* head encodes $a_j = \pi \tanh z_j$ by $R_y(a_j)$ on qubit j, applies $L = 4$ layers of trainable $R_y R_z$ rotations on every qubit, each followed by a ring of CNOTs $(q, q{+}1 \bmod 8)$ applied in order $q = 0, \ldots, 7$, measures $\langle Z_k \rangle$ on every qubit, and maps the eight expectations to ten logits with a linear layer (154 stored parameters, 146 of which can affect the output). Rotations are initialized in $\mathcal{U}(-0.1, 0.1)$. We simulate the circuit exactly with a batched PyTorch statevector simulator checked against PennyLane (Bergholm et al., 2018). Circuit variants change CNOT placement or rotation depth: *no CNOT*; a ring after the *last* layer only; $L \in \{1, 2\}$ layers; and disjoint *pairs* $(0,1), (2,3), (4,5), (6,7)$ after every layer or after the last layer only.

Protocol. For the small CNNs (the pretrained runs are specified in Section 5.6), training subsets are class-stratified samples of 250, 500 or 1,000 MNIST images and 1,000, 2,000 or 4,000 CIFAR-10 images; a disjoint stratified 2,000-image validation set selects the checkpoint epoch. Every run uses AdamW (weight decay 10^{-4}), a cosine schedule, batch size 32 and gradient clipping, for 50 epochs on MNIST and 80 on CIFAR-10 with standard augmentation. The backbone learning rate is half the head learning rate. For each dataset, size and head we select the head learning rate with the best mean validation accuracy over seeds 0–2 from a grid of half-decade steps, and report test accuracy once. Headline CIFAR-10/4,000 cells add seeds 3 and 4 at that selected rate. Heads in a cell share seeds, hence training subsets and validation splits, so we report paired differences (mean $\pm$ SD over seeds, in percentage points).

Common linear probe. To compare representations, we freeze each selected run's backbone, extract $\mathbf{z}$ on clean images, and fit a fresh linear layer $\mathbb{R}^8 \to \mathbb{R}^{10}$ (AdamW, cosine schedule, probe learning rate and epoch selected on validation). Its test accuracy, *probe accuracy*, compares the bottlenecks that the heads induced in joint training.

Zero-rotation identity. With every trainable rotation set to zero, the variational block reduces to its CNOTs. The encoded state is a product state with $\langle Z_j \rangle = \cos a_j$, and conjugating by $\mathrm{CNOT}(c, t)$ maps $Z_t \mapsto Z_c Z_t$ and leaves Z_c unchanged. Propagating Z_k back through the gates in reverse order, and cancelling repeated factors with $Z^2 = I$, gives a Pauli-Z string on a set S_k of qubits:

$$\langle Z_k \rangle\big|_{\theta=0} = \prod_{j \in S_k} \cos(\pi \tanh z_j), \qquad \bar{s} = \tfrac{1}{q} \textstyle\sum_k |S_k|. \tag{1}$$

We call $\bar{s}$ the *product order* ($q = 8$ qubits unless stated). It is 1 without CNOTs, 2.5 for the default circuit, 5.25 for a ring after the last layer or a single-layer circuit, 1.5 for pairs after the last layer, and 1 for pairs after every layer: the pair layer P satisfies $P^2 = I$, so its four copies cancel at zero rotations, while interleaved nonzero rotations generally prevent the cancellation (Figure 1a; Section A lists the supports). Without CNOTs the identity extends to all rotation angles: each qubit evolves alone, so

$$\langle Z_j \rangle = \alpha_j \cos a_j + \beta_j \sin a_j, \tag{2}$$

with α_j, β_j set by qubit j's trained rotations. With its readout layer, the ring-free head is thus a classical per-coordinate sinusoidal model. To test whether the products themselves carry the effect, we train classical heads on them with no circuit: *cos* uses $\cos a_j$ (order 1), *cos+sin* adds $\sin a_j$, and *cos-prod.* uses the products of Equation (1) for a given circuit's supports, each followed by a linear layer and tuned like every other head.

Table 1. Test accuracy (%) of three heads on the same CNN, each with its head learning rate selected by mean validation accuracy over the tested grid (backbone lr is coupled at $0.5\times$ the head lr); mean $\pm$ SD over 3 seeds; bold: highest mean in the row. MLP-8: one hidden layer of width 8; VQC: the default 8-qubit, 4-layer ring circuit. Differences are in percentage points, paired over seeds (same training subset and split). The last column repeats VQC $-$ MLP-8 with both heads at one shared head lr (10^{-3}). Every selected lr is interior to its grid.

		Selected lr			VQC $-$ (selected lr)		VQC $-$ MLP-8
Data	N_{train}	Linear	MLP-8	VQC	Linear	MLP-8	(lr 10^{-3})
MNIST	250	$\mathbf{90.3 \pm 1.9}$	78.7 ± 6.5	76.6 ± 5.3	-13.7 ± 5.9	-2.1 ± 9.7	$+10.8 \pm 13.9$
	500	$\mathbf{94.3 \pm 0.2}$	86.7 ± 3.0	90.0 ± 0.6	-4.3 ± 0.5	$+3.3 \pm 2.6$	$+6.1 \pm 11.6$
	1000	$\mathbf{96.4 \pm 0.5}$	93.9 ± 0.8	94.6 ± 0.4	-1.7 ± 0.9	$+0.8 \pm 1.0$	-0.5 ± 3.8
CIFAR-10	1000	$\mathbf{59.0 \pm 0.9}$	52.3 ± 1.8	43.1 ± 3.8	-15.8 ± 3.8	-9.2 ± 4.5	-13.6 ± 1.8
	2000	$\mathbf{67.6 \pm 0.7}$	62.0 ± 1.0	53.1 ± 0.9	-14.5 ± 1.3	-8.9 ± 1.7	-13.5 ± 2.0
	4000	$\mathbf{74.6 \pm 0.5}$	70.8 ± 0.5	67.7 ± 1.1	-7.0 ± 1.2	-3.1 ± 0.9	-10.1 ± 1.6

4. Tuning Changes Head Comparisons, but a Linear Head Stays Ahead

Before turning to representations, we check how much a head comparison depends on the training recipe. With one shared recipe (head learning rate 10^{-3}, 25 MNIST epochs; Section Q), the VQC beat MLP-8 on MNIST by up to 18 points and lost to it on CIFAR-10 by 8–14. Under the protocol of Section 3, these differences change in size and sign (Table 1): on MNIST, VQC $-$ MLP-8 becomes -2.1 ± 9.7, $+3.3 \pm 2.6$ and $+0.8 \pm 1.0$ at 250, 500 and 1,000 images, partly through the protocol itself and partly through per-head tuning (last column); on CIFAR-10 the deficit shrinks to 3.1–9.2 points. A linear head has the highest mean in all six settings and beats the VQC for every seed. Every selected rate is interior to its grid.

This audit fixes the comparisons used in the rest of the paper: every head is evaluated at its own validated learning rate. It also leaves a question. At 4,000 CIFAR-10 images the VQC trails the linear head by 7.0 ± 1.2 test points (5 seeds: 6.7 ± 1.0). Part of that gap could come from the VQC as a readout and part from the backbone it trains; a common probe measures the gap that remains under one readout.

5. What the VQC Head Does to the Backbone

5.1. The default VQC head lowers bottleneck probe accuracy

On CIFAR-10 with 4,000 images, the common probe reaches 69.8 ± 1.0 on backbones trained with the default VQC head and 74.5 ± 0.5 on backbones trained with a linear head, a paired difference of -4.7 ± 0.9 points that is negative for all five seeds. The probe reads the VQC-trained bottleneck better than the VQC head itself does (67.8 ± 1.3 test). The gap is larger with less data (-11.2 and -11.1 at 1,000 and 2,000 images) and smaller on MNIST (-7.9 to -1.0; 3 seeds each). A narrow classical head lowers probe accuracy as well: MLP-8 backbones probe 3.6 ± 0.6 points below linear-head backbones on CIFAR-10/4,000, 1.0 ± 0.9 above the VQC. The gap is not an artifact of linear decoding: a two-hidden-layer MLP probe on the same bottlenecks recovers at most 0.4 points more than the linear probe for any head, and puts the default VQC 4.8 ± 0.9 points below the linear head (Section F), also when its learning-rate grid is extended downward. The rest of this section shows that the VQC's share of the gap depends on its CNOT structure, which no classical head has.

5.2. Removing the CNOT rings recovers most of the gap

Deleting every CNOT and keeping the rest of the head unchanged raises probe accuracy to 73.6 ± 0.5 and test accuracy to 73.1 ± 0.5 (5 seeds). Against the default VQC this is $+3.8 \pm 1.2$ probe and $+5.2 \pm 1.4$ test points, positive for every seed, and the ring-free head now probes 2.8 ± 0.6 points above MLP-8. It leaves 0.8 ± 0.5 probe and 1.4 ± 0.5 test points to the linear head. On MNIST, ring removal gains 2.8 and 1.4 probe points at 500 and 1,000 images.

5.3. Product order tracks the gap

Which property of the rings matters? Figure 1b places every head at the product order of Equation (1). Across the eleven ring-family heads, at four orders, every head's mean probe accuracy is below that of every lower-order head. Depth and parameter count do not explain this: a ring after the last layer keeps the default's parameter count and rotation layers but raises the order to 5.25 and lowers the probe by 9.7 ± 2.2 points (5 seeds), and the 1- and 2-layer circuits at that order probe at 56.8 and 61.2 (3 seeds).

Order also carries over to layouts it was not fit on. For six CNOT layouts outside the ring family, chosen by order alone, we predicted probe accuracy from the ring-family fit before training. The association holds (Spearman -0.89, against a threshold of -0.8 set in advance; five of six within 4 points), with two rank reversals. The largest miss, brickwork at 5 points above its prediction, is the layout trained furthest from its product model. Order extends beyond zero rotations: expanding each readout operator

Table 2. Order without CNOTs, without entanglement and without a saturating encoding (CIFAR-10, 4,000 images, small CNN, 5 seeds; pre-registered). Probe: common linear probe (mean ± SD). Fit: least-squares line $76.6-3.1\,\bar{s}$ over the four ring-family layouts (orders 1, 1.5, 2.5, 5.25), committed before these runs with one layout then at 3 seeds (the five-seed refit agrees at this precision; cf. the three-seed fit of Figure 1b). CZ commutes with Z, so a CZ ring entangles but has order 1; $\langle Z_k Z_{k+1}\rangle$ and $\langle Z_k Z_{k+1} Z_{k+2}\rangle$ on the CNOT-free circuit are exact products of 2 or 3 per-coordinate sinusoids at any rotation. Bottom row: encoding $a_j = z_j$, which does not saturate, on the same four layouts; slope -2.1 points per unit order, against -3.1 under $\pi\tanh$.

Head	Entangles	$\bar{s}$	Probe	Fit
VQC, no CNOT	no	1	73.6 ± 0.5	73.5
VQC, CNOT ring	yes	2.5	69.8 ± 1.0	68.9
VQC, CZ ring	yes	1	73.2 ± 0.9	73.5
no CNOT, $\langle Z_k Z_{k+1}\rangle$	no	2	70.3 ± 1.5	70.4
no CNOT, $\langle Z_k Z_{k+1} Z_{k+2}\rangle$	no	3	67.4 ± 1.8	67.3
$a_j = z_j$, four layouts	varies	1–5.25	72.5, 71.7, 69.8, 63.7	

$U(\theta)^\dagger Z_k U(\theta) = \sum_P c_P P$ in Pauli strings, the *effective order* is the c_P^2-weighted mean number of inputs touched by the strings visible on the encoded state, which equals $\bar{s}$ at $\theta = 0$ (Section N). At its trained rotations brickwork's effective order is 2.6 rather than 3.5, below the distance-2 ring's 3.0 (Figure 1b, Section G). Under $\frac{\pi}{2}\tanh z + \frac{\pi}{2}$ six layouts follow order as predicted (-0.94); under a phase encoding, whose zero-rotation readouts vanish, probe accuracy follows the circuits' R_y order (-0.99, Sections H and I). At 12 qubits the ordering holds as predicted, with the default ring again 4.0 points below the ring-free one (Section J).

Two interventions test the products directly. First, classical heads trained end to end on a circuit's zero-rotation products, with no circuit, land at the circuit's order on the trend: the products of the default circuit give 68.8 ± 1.6 probe and 67.1 ± 1.3 test, paired differences to the default VQC of -1.0 ± 2.2 and -0.7 ± 1.7 with mixed signs across seeds. Summing instead of multiplying the same cosines would give the order-1 cos head's function class (73.3 probe), as every CNOT-derived support matrix has full rank, though such a head would train differently. Second, the *pairs* circuit keeps a CNOT after every layer but has order 1, because its pair CNOTs cancel at zero rotations. It recovers 3.1 ± 1.2 probe and 4.4 ± 1.2 test points over the default, positive for every seed, and sits 0.7 ± 1.0 probe points below the ring-free circuit. Pairs after the last layer only (order 1.5) probe at 71.2 ± 0.7, between orders 1 and 2.5, as does the classical head on their products (71.2 ± 1.3).

Order needs neither CNOTs, entanglement nor a saturating encoding (Table 2). A CZ ring entangles, but CZ commutes with Z, so its order is 1: it probes 3.4 ± 0.8 points above the default CNOT ring (every seed) and 0.4 ± 0.8 below the ring-free head. Conversely, two- and three-qubit Z readouts of the ring-free circuit, which has no entangling gate, probe at 70.3 and 67.4, against 70.4 and 67.3 from the ring-family fit. Under the unbounded encoding $a_j = z_j$ four layouts keep their order ranking.

In CNOT-derived supports, order rises with overlap, the number of inputs two readouts share. Classical product heads on designed supports separate the two: at equal overlap and full rank an order-3 star probes 5.4 ± 2.7 points above an order-4 layout (5 seeds, all positive), and at equal order how the coupling is distributed also matters (Table 8). Order tracks lower mean probe accuracy but does not determine it.

The pattern replicates on MNIST with 1,000 images under an ordering logged before the runs but committed after them (Section U): probe accuracy is 96.5, 95.5, 95.1 and 91.6 at orders 1, 1.5, 2.5 and 5.25 (Section C).

The identity is guaranteed at zero rotations and generally fails at trained ones. An affine map from each trained circuit's zero-rotation readouts to its trained readouts recovers 87% and 90% of the held-out readout variance for the default and last-ring circuits (Section B), and the cost does not depend on starting in the product regime: with rotations initialized in $\mathcal{U}(-\pi, \pi)$ the default circuit still probes 4.6 ± 1.7 points below the ring-free one (Table 14).

5.4. Without CNOTs the head is a classical sinusoidal model

By Equation (2), the ring-free head computes $W[\alpha_j \cos a_j + \beta_j \sin a_j]_j + b$ for every setting of its rotations. Table 3 compares it with classical heads of this form trained under the same protocol. The cos-only restriction trails it by 0.3 ± 0.6 probe and 0.6 ± 0.7 test points with mixed signs, and the unrestricted cos+sin extension by 0.9 and 1.3 (3 seeds). The match is of function class, not parameters (Section B).

5.5. Controls that do not recover the gap

Table 10 (Section E) tests alternative explanations on CIFAR-10/4,000. Lowering or raising the backbone learning rate with the head learning rate fixed leaves a gap of 4.9–8.6 points at every rate where the VQC trains, and the VQC collapses at the two highest rates. Freezing the circuit's rotations at initialization does not recover the probe (-1.6 ± 0.7 against the trained circuit). BatchNorm before the encoding and two alternative encodings, one of which breaks the sign symmetry of $\cos(\pi\tanh z)$, recover at most 0.7 probe points.

Higher-order circuits also fit their training data less well under the 80-epoch budget (clean training accuracy 96.6% without CNOTs, 89.2% for the default circuit and 65.6% for a ring after the last layer), so part of the gap could be under-training. Doubling the budget to 160 epochs for

Table 3. Without entangling gates (and with single-qubit Z readouts) the VQC head is exactly a classical per-coordinate sinusoidal model, $\text{logits} = W[\alpha_j \cos a_j + \beta_j \sin a_j]_{j=1}^{8} + b$ with $a_j = \pi \tanh z_j$ and α_j, β_j set by the trained rotations, for all rotation angles. Classical heads $W[\cos a_j]_j + b$ (a restriction) and $W[\cos a_j, \sin a_j]_j + b$ (an unrestricted extension), trained end to end with the same recipe and per-head validation lr selection, trail it by 0.3–0.9 probe points and 0.6–1.3 test points. CIFAR-10 with 4,000 training images; test accuracy of the full model and of the common linear probe on the bottleneck (%, mean ± SD over n seeds); differences in percentage points, paired over seeds.

				− ring-free VQC	
Head	n	Test	Probe	Test	Probe
VQC, no CNOT rings	5	73.1 ± 0.5	73.6 ± 0.5	—	—
cos-only (restriction)	5	72.5 ± 0.5	73.3 ± 0.4	-0.6 ± 0.7	-0.3 ± 0.6
cos+sin (unrestricted extension)[a]	3	72.0 ± 0.2	72.8 ± 0.4	-1.3 ± 0.4	-0.9 ± 0.5
linear (reference)	5	74.5 ± 0.7	74.5 ± 0.5	$+1.4 \pm 0.5$	$+0.8 \pm 0.5$

[a] 3 seeds; differences are against the ring-free VQC on the same seeds (test 73.3, probe 73.7).

Table 4. The representation cost on an ImageNet-pretrained ResNet-18 fine-tuned through each head (CIFAR-10; backbone learning rate $0.1\times$ the head's; 30 epochs at 128 px). Probe: test accuracy (%) of the common linear probe on the frozen 8-dimensional bottleneck, and its paired difference to the linear head in percentage points (mean ± SD; 4,000 images: number of seeds in parentheses; 1,000 and 10,000 images: 3 seeds). Head learning rates are selected on validation separately at each size. At 4,000 images every head's grid spans 3×10^{-4}–10^{-2}, extended to 10^{-4} for the default and last-ring circuits; at 10,000 images it spans 3×10^{-4}–3×10^{-3}, extended to 10^{-4} for the linear, default and last-ring heads. At both sizes every selected rate is interior. The linear row gives its absolute probe accuracy at 1,000 and 10,000 images. The last column repeats the 4,000-image difference with a two-hidden-layer MLP probe (3 seeds).

		1,000 images	4,000 images		10,000 images	4,000, MLP probe
Head	Order	− linear	probe	− linear	− linear	− linear
linear	—	82.9 ± 0.9	89.4 ± 0.4 (5)	—	92.6 ± 0.4	—
VQC, no CNOT	1	-1.6 ± 1.2	89.4 ± 0.4 (5)	0.0 ± 0.1	-0.1 ± 0.4	-0.6 ± 0.3
VQC, pairs	1		89.3 ± 0.4 (5)	-0.2 ± 0.3		-0.8 ± 0.5
cos-prod. (default supports)	2.5		85.5 ± 1.6 (5)	-4.0 ± 1.6		-3.0 ± 1.1
VQC, ring (default)	2.5	-22.3 ± 2.9	82.3 ± 2.3 (5)	-7.2 ± 2.3	-1.9 ± 0.4	-5.8 ± 0.3
VQC, ring, last	5.25	-34.1 ± 6.3	71.4 ± 4.8 (5)	-18.0 ± 4.7	-8.0 ± 1.0	-18.5 ± 3.9

the linear, default, ring-free and last-ring heads (CIFAR-10/4,000) raises every probe but keeps the ordering (Table 9). The default circuit then fits its training set nearly as well as the ring-free and linear heads (98.8% vs 99.9%) and still probes 2.8 ± 0.4 points below the ring-free head and 3.1 ± 0.3 below the linear one (5 seeds, every seed negative). The last-ring circuit remains under-fit at 160 epochs (79.5%), so its larger cost includes slower optimization.

5.6. A pretrained backbone, data size and training budget

We repeat the key heads on an ImageNet-pretrained ResNet-18 fine-tuned through each head, its classifier replaced by the 8-dimensional bottleneck (Table 4). The ordering carries over: at 4,000 images the ring-free and pairs circuits probe within 0.2 points of the linear head, the default circuit 7.2 ± 2.3 below it and the last-ring circuit 18.0 ± 4.7 below (5 seeds, every seed negative). At a fixed 30 epochs the cost falls with training size, from 22.3 ± 2.9 points at 1,000 images to 1.9 ± 0.4 at 10,000, but smaller sets then also get fewer optimizer updates. At an approximately matched number of updates (about 3,800) the default circuit costs 6.6 ± 1.7, 5.7 ± 0.7 and 7.2 ± 2.1 points at 1,000, 4,000 and 10,000 images, and the ring-free head stays within 0.5 points of linear (seeds 0–2; Table 24); re-selecting the learning rates gives similar costs. Training budget therefore accounts for much of the size dependence, though the two are not fully separated.

High-order heads are unstable here (Section O); pre-registered tests keep the ordering in every seed (Section P): on SVHN (10 seeds), with data re-uploading, and on a pre-trained MobileNetV3-Small, where the cost is small (default 1.6 ± 0.8 below linear) and one criterion misses.

5.7. Why order costs: a signal budget and a saturation trap

Differentiating Equation (1) at zero rotations gives, for $j \in S_k$,

$$\frac{\partial \langle Z_k \rangle}{\partial z_j} = -\pi \operatorname{sech}^2(z_j) \sin a_j \prod_{i \in S_k \setminus \{j\}} \cos a_i, \qquad (3)$$

so a coordinate's gradient is scaled by the cosines of the other coordinates in S_k (Proposition 2) and by its own $\operatorname{sech}^2(z_j)$. A readout also keeps a useful magnitude only if its factors stay near ± 1, a signal budget:

Table 5. Releasing or penalizing bottleneck saturation recovers part of the cost at high order (CIFAR-10, 4,000 images, small CNN, 5 seeds; pre-registered). Probe: unregularized head (mean); Δ esc., Δ sat., Δ dec.: paired probe difference to it (mean ± SD) with the escape-only straight-through estimator (forward pass unchanged; mean $S \leq 0.005$ in every head), the anti-saturation and the decorrelation penalty. S: fraction of saturated bottleneck coordinates ($|\tanh z| > 0.9$), unregularized → anti-saturation. Replications on seeds 5–9 are in the text and Sections K and M. "prod.": classical heads on the circuits' zero-rotation products. Details in Section K.

Head (order)	Probe	Δ esc.	Δ sat.	Δ dec.	S
VQC, no CNOT (1)	73.6	$+0.9 \pm 0.5$	$+0.6 \pm 0.8$	—	0.01→0.00
VQC, ring (2.5)	69.8	$+1.5 \pm 0.9$	$+1.7 \pm 1.3$	—	0.30→0.00
VQC, last ring (5.25)	60.1	$+6.8 \pm 2.2$	$+7.9 \pm 3.0$	—	0.49→0.00
prod., cos (1)	73.3	—	$+0.2 \pm 1.1$	-0.3 ± 0.4	0.05→0.02
prod., ring (2.5)	68.8	$+2.7 \pm 1.8$	$+2.2 \pm 1.7$	$+0.3 \pm 1.9$	0.41→0.03
prod., last ring (5.25)	54.8	$+11.3 \pm 4.5$	$+12.6 \pm 4.9$	$+7.8 \pm 4.1$	0.60→0.03

Proposition 1 (Signal budget)**.** *Let* $r_k = \prod_{j \in S_k} \cos a_j$ *with* $|S_k| = m$*,* $a_j = \pi \tanh z_j$ *and independent factors, and let* $\mathbb{E}|r_k| \geq a > 0$*. Then some* $j \in S_k$ *has* $\Pr\left(\delta < |\tanh z_j| < 1 - \delta\right) \leq (1 - a^{1/m})/(1 - \cos \pi\delta)$ *for every* $\delta \in (0, \frac{1}{2})$*; if the factors are identically distributed, every* $j \in S_k$ *does. Without independence, every* $j \in S_k$ *satisfies the bound with* a *in place of* $a^{1/m}$*.*

As m grows, $a^{1/m} \to 1$, so high-order inputs gather at $z \approx 0$ or at saturation. On the trained bottlenecks the middle-zone fraction ($0.1 < |\tanh z| < 0.9$) is 0.84, 0.63 and 0.46 at orders 1, 2.5 and 5.25, and the lost mass goes to saturation (30–49% of coordinates for the two ring circuits, 1% without CNOTs), which tracks probe accuracy across the 28 VQC configurations ($\rho = -0.91$; Table 20). On these bottlenecks the products keep their magnitude (mean $\mathbb{E}|r_k|$ 0.61 to 0.44; the decoder does not compensate), but at these orders the bound is loose: it gives the direction of the depletion, not its size (Section L).

Both ends of the encoding are flat (Proposition 3): with $t = |\tanh z|$, $|\frac{d}{dz} \cos(\pi \tanh z)| \leq \pi^2 t$ near zero and $\leq 8\pi^2 e^{-4|z|}$ at saturation, the square of tanh's decay, because $\sin a_j$ vanishes there too (Proposition 3 bounds the resulting loss of sensitivity). Every head saturates early in training ($S = 0.44$–0.77 at epoch 5, 3 seeds); the ring-free head then desaturates to 0.02, while the default and last-ring heads stall at 0.30 and 0.46 and their probe gaps peak at epochs 15 and 35 (Section M). We tested this trap with a pre-registered intervention that leaves the forward pass unchanged: our escape-only variant of the straight-through estimator (Bengio et al., 2013), which in the backward pass drops $\operatorname{sech}^2(z_j)$ when the descent step moves z_j toward 0, so saturated coordinates can leave (a plain version made every circuit diverge, and we redesigned it after; Section M). It removes saturation in every head (mean $S \leq 0.005$) and raises probe accuracy by 6.8 ± 2.2 points on the last-ring circuit and 1.5 ± 0.9 on the default, positive in every seed, against 0.9 ± 0.5 without CNOTs (Table 5). The default gain misses the committed $+1.5$ by 0.02; a pre-registered replication on seeds 5–9 meets every criterion ($+3.3$, $+5.6$ and $+1.3$; Section M). A penalty against saturation, which also changes the loss, gives similar gains (Section K).

With saturation removed, order still costs: the escape-only circuits probe at 74.5, 71.3 and 66.9 at orders 1, 2.5 and 5.25. Releasing the trap recovers half of the last-ring cost and 1.5 of the default's 3.8 points (44% and 15% if both arms are treated); the rest is consistent with the budget, though not uniquely attributed to it.

6. Discussion and Limitations

Implications for hybrid design. For CNN-fed quantum heads trained end to end, the entangling layout and the observables are choices about the backbone: product order, computable before training, ranks them, with a few reversals. The best quantum heads are low-order: the ring-free head (exactly classical, Equation (2)) or the pairs circuit.

Limitations. We study two families of CNOT layouts and a CZ ring (8 qubits, and 12 for the headline heads; $R_y R_z$ layers, angle encodings, mainly $\pi \tanh$), one qubit per bottleneck coordinate, in noiseless simulation, with small CNNs and two pretrained networks on subsets of MNIST, CIFAR-10 and SVHN of at most 10,000 images; the cost varies across pretrained backbones (1.6–7.2 points on CIFAR-10/4,000) and is untested with non-angle encodings or on hardware. Most cells use 3 seeds (5 or 10 where stated), too few for equivalence claims. Product order is a coarse scalar: it misses brickwork by 5 points, its cost per unit does not carry across widths, and at fixed order the distribution of coupling also matters (Table 8; classical heads only). The identity is guaranteed at zero rotations, and trained circuits on the small CNNs move away from it (held-out R^2 0.51–0.90). The signal budget assumes independent factors, is loose at these orders and is checked only at the end of training; whether it mediates the cost, and how it and the saturation trap divide the cost at low order, are not settled. Information recoverable only by decoders stronger than a two-hidden-layer MLP is not measured.

7. Conclusion

A jointly trained VQC head changes what the CNN learns: the default head costs 4.7 CIFAR-10 probe points. Across layouts, encodings, a CZ ring and multi-qubit readouts the cost tracks product order, not the presence of entangling gates. For independent factors that keep their product's magnitude, high order empties the middle of the encoding; our high-order bottlenecks saturate early and stay, and releasing them recovers half the top-order cost.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, specifically the evaluation of hybrid quantum–classical classifiers. Its practical consequence is methodological: comparisons between quantum and classical heads should tune each head separately and measure the representation the head induces. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. The Zero-Rotation Identity

Let $U(\theta)$ be the variational block and $|\psi(\mathbf{z})\rangle = \bigotimes_j R_y(a_j)\,|0\rangle$ the encoded product state, so that $\langle Z_k\rangle = \langle\psi|\,U^\dagger Z_k U\,|\psi\rangle$. At $\theta = 0$ every R_y and R_z is the identity and U is the product of the circuit's CNOTs. Conjugation by $\mathrm{CNOT}(c,t)$ maps $Z_t \mapsto Z_c Z_t$ and fixes Z_c, so $U^\dagger Z_k U$ is obtained by applying these rules to Z_k for the gates in reverse order; the result is a string $\prod_{j\in S_k} Z_j$, with S_k the set of qubits on which an odd number of propagation paths land. Since $|\psi\rangle$ is a product state with $\langle\psi|\,Z_j\,|\psi\rangle = \cos a_j$, the expectation factorizes into Equation (1). Table 6 lists the supports. Pairs after every layer apply each pair CNOT four times, which is the identity, so their supports equal the no-CNOT ones; with trained rotations between the layers the gates no longer cancel (its outputs differ from the ring-free circuit's at random rotation angles), and the circuit can entangle: with every input at zero and every rotation zero except $R_y(\pi/2)$ on qubit 0 in the last layer, the final pair CNOT turns qubits 0 and 1 into a Bell pair, whose single-qubit reduced state has purity $1/2$.

For the ring-free circuit and any rotation angles, qubit j starts from Bloch vector $(\sin a_j, 0, \cos a_j)$ after encoding and then undergoes a fixed rotation $M_j \in \mathrm{SO}(3)$ determined by its trained R_yR_z layers. Hence $\langle Z_j\rangle = (M_j)_{zx}\sin a_j + (M_j)_{zz}\cos a_j$, which is Equation (2) with $\beta_j = (M_j)_{zx}$ and $\alpha_j = (M_j)_{zz}$. The test suite checks the formula against the simulator (absolute tolerance 10^{-10}, relative 10^{-5}). Since the readout layer is linear, the ring-free head is $W[\alpha_j\cos a_j + \beta_j \sin a_j]_j + b$.

Table 6. Zero-rotation supports S_k of each readout $\langle Z_k\rangle$, written as digit strings (e.g. 015 is $\{0,1,5\}$), and the mean product order $\bar{s}$, computed by propagating Z_k backwards through each circuit's CNOTs.

Circuit	$\bar{s}$	S_0	S_1	S_2	S_3	S_4	S_5	S_6	S_7
no CNOT	1	0	1	2	3	4	5	6	7
pairs (every layer)	1	0	1	2	3	4	5	6	7
pairs, last	1.5	0	01	2	23	4	45	6	67
ring (default)	2.5	015	256	367	047	15	26	37	04
ring, last	5.25	1234567	01	012	0123	01234	012345	0123456	01234567
ring, $L = 1$	5.25	1234567	01	012	0123	01234	012345	0123456	01234567
ring, $L = 2$	5.25	01357	0234567	134567	024567	13567	02467	1357	0246

B. Trained Circuits against the Product Model

The identity is guaranteed at zero rotations and generally fails at trained ones, so we fit an affine map from each trained circuit's zero-rotation readouts to its trained readouts on half of the validation set and evaluate it on the other half (Table 7; a frozen circuit gives $R^2 = 0.997$). For the default and last-ring circuits it recovers 87% and 90% of the held-out readout variance, and the bottleneck gradient through the remapped products aligns with the trained one (median cosine 0.86 and 0.87). The products miss some task detail: the trained readout on the remapped products scores 7 and 6 points lower. Low-order circuits move further (R^2 0.73–0.83; for the ring-free circuit through the $\sin a_j$ term of Equation (2)). At order 5.25 the trained rotations may recover part of the cost: the last-ring circuit tests 6.4 ± 5.8 points above the classical head on its products (four of five seeds; 95% interval $[-0.8, 13.6]$). The cost does not depend on starting in the product regime: with the rotations initialized in $\mathcal{U}(-\pi,\pi)$, where the identity does not hold at the start, the default circuit still probes 4.6 ± 1.7 points below the ring-free one (3.8 ± 1.6 on the same three seeds at the default initialization), and neither circuit's probe moves by more than a point on average (Table 14); each default readout then depends on all eight inputs (Section N).

Table 7 compares each trained circuit with the same circuit after setting every rotation to zero, on the validation embeddings of the run's own backbone. The frozen circuit anchors the measurement: rotations held at their $\mathcal{U}(-0.1, 0.1)$ initialization reproduce the product model almost exactly. Held-out R^2 equals the in-sample value to within 0.015 for every run in this table, so the affine map does not overfit. These are endpoint measurements: they say how close the trained circuit is to its product model, not that training followed the product model's trajectory. Two results bound the approximation. First, on the small CNN the products, even after remapping, lose task-relevant detail: held-out accuracy drops by 6–12 points (by under 1 point on the pretrained backbone). Second, gradient alignment depends on the remapping: with the trained readout applied directly to the zeroed circuit, the median loss-gradient cosine is 0.59 for the default circuit, and -0.08 for its seed 4 (0.96 after remapping); after remapping it is 0.79–0.87 for every trained entangling small-CNN circuit in this table (0.91–0.98 on the pretrained backbone), with 3–7% of samples negative on average (18% in the worst run). Zeroing the rotations without remapping costs 14–23 accuracy points on average for the two ring circuits and 31–43 for the pairs and CNOT-free circuits, which mostly reflects the unrefitted readout. Low-order circuits drift further, as Section 5.4 explains for the ring-free case.

Table 7. Trained circuits against their own zero-rotation product model (CIFAR-10, 4,000 training images; small CNN, and below the pretrained ResNet-18; each run's validation embeddings, split by index parity; means over n seeds). An affine map from the readouts of the same circuit with all rotations set to zero, q_0, to the trained readouts q is fit on one half and evaluated on the other. R^2: held-out fraction of trained readout variance it recovers, a ratio relative to the circuit's own output spread; MSE: the unnormalized held-out mean squared error per readout ($\langle Z \rangle \in [-1, 1]$). Accuracy: held-out accuracy of the trained head, and of its trained readout layer applied to the remapped q_0. Loss-gradient cosine: median over held-out samples of the cosine between $\partial\mathcal{L}/\partial z$ (cross-entropy) of the trained head and of the zeroed head with the trained readout kept (zeroed) or applied to the remapped q_0 (remapped). Last column: mean $|\theta|$ of the R_y / R_z rotations (rad).

					Accuracy		Loss-grad. cosine		
Circuit	Order	n	R^2	MSE	trained	remapped	zeroed	remapped	$\lvert R_y\rvert$ / $\lvert R_z\rvert$
ring (default), frozen at init	2.5	3	0.997	0.0010	67.6	67.0	1.00	1.00	0.05 / 0.05
ring (default)	2.5	5	0.866	0.0414	69.3	62.3	0.59	0.86	0.14 / 0.48
ring, last	5.25	5	0.903	0.0326	55.7	49.4	0.75	0.87	0.10 / 0.41
pairs, last	1.5	5	0.826	0.0684	70.0	60.4	0.49	0.81	0.20 / 0.37
pairs	1	5	0.788	0.0990	73.9	64.3	0.42	0.79	0.32 / 0.43
no CNOT	1	5	0.734	0.1309	74.1	61.9	0.34	0.71	0.41 / 0.28
Pretrained ResNet-18 backbone									
ring (default)	2.5	5	0.997	0.0019	78.2	78.1	0.98	0.98	0.02 / 0.05
ring, last	5.25	5	0.994	0.0037	58.6	59.3	0.91	0.91	0.05 / 0.05
no CNOT	1	5	0.992	0.0067	90.4	90.2	0.93	0.93	0.06 / 0.05

C. Order versus Overlap, and Training Budget

The pattern also replicates on MNIST with 1,000 images, where we fixed the layouts and wrote the predicted ordering in our log before running them, though it was first committed after them (Section U). Probe accuracy is 96.5 without CNOTs, 95.5 for pairs after the last layer, 95.1 for the default circuit and 91.6 for a ring after the last layer; the small pairs-last advantage over the default is $+0.4 \pm 0.5$ (5 seeds, 4 positive); the classical head on the default products probes 0.1 ± 0.5 below the default VQC, and the one on the last-ring products falls below that circuit (90.4).

In CNOT-derived supports, order rises with overlap, the number of inputs two readouts share, so classical product heads on designed supports separate the two (Table 8). At equal overlap and full rank, the order-3 star probes 5.4 ± 2.7 points above an order-4 layout (5 seeds, all positive); the other matched comparisons point the same way but also change rank or match overlap only approximately. At equal order the stars, which reuse one or two inputs across most products, probe 3.0 and 1.8 points above the windows (the latter with mixed signs), so how the coupling is distributed also matters, and an order-3 star probes as high as the default ring's order-2.5 products (68.9 vs 68.8). The designed heads also differ in training fit (85.6% vs 74.1% clean training accuracy for the order-3 star and the order-4 layout), so their contrasts mix structure with optimization. Within the tested family and protocols, higher order tracks lower mean probe accuracy, but order is not sufficient to determine it.

Table 8 lists the designed-support heads of Section 5.3. Overlap is the mean over ordered readout pairs $k \neq l$ of $|S_k \cap S_l|$; if input j appears in d_j supports, order is $\frac{1}{8}\sum_j d_j$ and overlap is $\frac{1}{56}\sum_j d_j(d_j - 1)$, so for windows, where every d_j equals the order, the two rise together. Stars hold the order while concentrating the d_j on one or two inputs. Windows of width 2 and 4 have rank-deficient support matrices; the gapped window of width 4 restores full rank at the same order and overlap and probes 0.2 ± 4.1 points above the rank-5 window (mixed signs), so restoring full rank does not remove the order-4 deficit. Table 9 gives the training-budget check of Section 5.5, with the linear head also rerun at 160 epochs.

D. Gradient Attenuation at Zero Rotations

Proposition 2. *At zero rotations, for $j \in S_k$ and $a_i = \pi \tanh z_i$,*

$$\left|\frac{\partial\langle Z_k\rangle}{\partial z_j}\right| \leq \pi \prod_{i \in S_k \setminus \{j\}} |\cos a_i|.$$

If, in addition, the coordinates z_i, $i \in S_k$, are independent, then

$$\mathbb{E}\Big[\Big(\frac{\partial\langle Z_k\rangle}{\partial z_j}\Big)^2\Big] \leq \pi^2 \prod_{i \in S_k \setminus \{j\}} \mathbb{E}\big[\cos^2 a_i\big] \leq \pi^2 \rho^{|S_k|-1}, \qquad \rho = \max_{i \in S_k} \mathbb{E}\big[\cos^2 a_i\big] \leq 1.$$

Table 8. Classical product heads whose supports separate product order from overlap (mean pairwise $|S_k \cap S_l|$), trained end to end like every other head (CIFAR-10, 4,000 training images; n seeds, each at its validation-selected learning rate; differences paired over shared seeds). Rank: rank of the 8×8 support matrix, which bounds the number of independent input directions the products can respond to. Window w: $S_k = \{k, \ldots, k+w-1\}$; gapped window 4: $\{k, k+1, k+2, k+4\}$ (mod 8). Star 2: $\{1,2\}, \{0,1\}, \{0,2\}, \ldots, \{0,7\}$. Star 3: $\{1,2,3\}, \{0,1,2\}, \{0,1,3\}, \ldots, \{0,1,7\}, \{0,2,3\}$. Bottom: paired probe differences and what each comparison holds fixed.

Supports	Order	Overlap	Rank	n	Test	Probe
cos (order 1)	1	0.00	8	3	72.2 ± 0.3	73.2 ± 0.1
window 2	2	0.29	7	3	68.1 ± 0.4	69.3 ± 0.4
star 2	2	0.82	8	5	69.8 ± 2.2	70.9 ± 1.9
default ring products	2.5	0.57	8	3	66.3 ± 0.9	67.8 ± 1.1
window 3	3	0.86	8	5	64.2 ± 2.9	67.1 ± 2.3
star 3	3	1.71	8	5	67.2 ± 1.2	68.9 ± 1.1
window 4	4	1.71	5	5	60.1 ± 3.9	63.3 ± 3.1
gapped window 4	4	1.71	8	5	61.0 ± 2.4	63.5 ± 1.8

Comparison	Held fixed	Probe difference
star 3 − gapped window 4	equal overlap, both rank 8	$+5.4 \pm 2.7$
star 2 − window 3	approximately equal overlap, both rank 8	$+3.9 \pm 1.9$
star 3 − window 4	equal overlap, rank 8 vs 5	$+5.6 \pm 2.2$
star 2 − window 2	equal order, rank 8 vs 7	$+3.0 \pm 0.2$
star 3 − window 3	equal order, both rank 8	$+1.8 \pm 2.8$

Table 9. Twice the training budget (160 instead of 80 epochs, each head at its 80-epoch learning rate and cosine schedule; CIFAR-10, 4,000 training images; n seeds). Clean-train: accuracy on the unaugmented training subset at the selected checkpoint; loss: final-epoch training loss on augmented batches. At 160 epochs, default − no-CNOT probe accuracy is -2.8 ± 0.4 and default − linear is -3.1 ± 0.3 (paired).

Head	Epochs	n	Clean-train	Train loss	Test	Probe
linear	80	5	98.7 ± 0.4	0.13	74.5 ± 0.7	74.5 ± 0.5
	160	5	99.9 ± 0.1	0.03	75.6 ± 0.6	75.5 ± 0.4
no CNOT	80	5	96.6 ± 0.7	0.22	73.1 ± 0.5	73.6 ± 0.5
	160	5	99.9 ± 0.1	0.03	74.6 ± 0.5	75.1 ± 0.4
ring (default)	80	5	89.2 ± 2.5	0.44	67.8 ± 1.3	69.8 ± 1.0
	160	5	98.8 ± 0.6	0.08	70.6 ± 0.5	72.4 ± 0.6
ring, last	80	5	65.6 ± 5.4	1.02	55.3 ± 3.1	60.1 ± 3.1
	160	3	79.5 ± 15.1	0.67	59.6 ± 7.6	64.7 ± 4.7

Proof. Differentiating Equation (1) gives $|\partial \langle Z_k \rangle / \partial z_j| = \pi \operatorname{sech}^2(z_j) \, |\sin a_j| \prod_{i \in S_k \setminus \{j\}} |\cos a_i|$, and $\operatorname{sech}^2 \leq 1$, $|\sin| \leq 1$. Under independence the expectation of the squared product factorizes, and $\mathbb{E}[\pi^2 \operatorname{sech}^4(z_j) \sin^2 a_j] \leq \pi^2$. □

Under independence, each additional input in a readout's support therefore multiplies the bound on the mean squared gradient that every other input receives through that readout by $\mathbb{E}[\cos^2 a_i]$, which is below 1 unless a_i concentrates at 0 or $\pm\pi$; if $\rho = 1$ the bound need not fall. The proposition bounds the size of the per-readout gradient; it does not by itself show that the backbone learns a worse representation, and an input can still receive gradient through other readouts. On the trained bottlenecks (CIFAR-10, 4,000 images, seeds 0–2, validation embeddings), $\mathbb{E}[\cos^2 a_j]$ ranges from 0.37 to 1.0 across coordinates (mean 0.63 for the default circuit, 0.72 for the last-ring circuit), and the measured $\mathbb{E}[\prod_i \cos^2 a_i]$ follows the independence product closely: 0.39 vs 0.39 with two other factors for the default circuit, and 0.11 vs 0.09 with seven for the last-ring circuit, a factor of roughly 0.6–0.8 per additional input. Coordinates with $\mathbb{E}[\cos^2 a_j]$ near 1 sit at $a_j \approx \pm\pi$, where $\tanh$ saturates: they leave the other inputs' gradients unattenuated but receive almost none themselves.

E. Controls

Table 10 gives the controls summarized in Section 5.5.

Table 10. Controls on CIFAR-10 with 4,000 training images. Entries are differences in the test accuracy of the common linear probe on the frozen 8-dimensional bottleneck, in percentage points, paired over 3 seeds (mean ± SD). The five controls leave a substantial probe gap; removing the CNOT rings recovers most of it.

Control	Comparison	Probe Δ	Observation
Reference: default VQC	VQC − linear	-4.7 ± 1.2	the gap under study
Backbone lr[a]	VQC − linear, backbone lr 0.0025	-6.3 ± 1.2	gap persists at every tested backbone lr where the VQC trains
	VQC − linear, backbone lr 0.005	-4.9 ± 1.2	
	VQC − linear, backbone lr 0.015	-8.6 ± 0.5	
Freeze circuit at init	frozen − trained circuit	-1.6 ± 0.7	gap persists with rotations frozen at initialization
BatchNorm before encoding	VQC+BN − VQC	-1.2 ± 2.2	pre-encoding BN does not recover the probe
	VQC+BN − linear	-5.9 ± 3.4	
Monotone encoding[b]	monotone − default	$+0.3 \pm 0.7$	neither encoding recovers more than 0.7 points
	symmetric control − default	$+0.7 \pm 0.5$	
	monotone − symmetric control	-0.4 ± 0.9	
Contrast: remove CNOT rings	ring-free − default	$+3.8 \pm 1.6$	recovers most of the gap (§5.2)
	ring-free − linear	-0.8 ± 0.7	

[a] Head lr fixed at 0.01 (VQC) and 0.03 (linear) while the backbone lr varies; the VQC collapses at the tested backbone lrs 0.03 and 0.06. Other rows: each head at its validation-selected head lr, backbone lr $0.5\times$ head lr. [b] Monotone: $R_y(\frac{\pi}{2}\tanh z + \frac{\pi}{2})$; symmetric control: $R_y(\frac{\pi}{2}\tanh z)$, same range width; default: $R_y(\pi \tanh z)$. Not a probe difference: the final-epoch pre-clip backbone gradient norm is larger with the VQC head than with the linear head (1.92 ± 0.13 vs 0.75 ± 0.02).

F. Nonlinear Probe, Gradient Rank, Initialization and Gradient Decoupling

Table 11 refits the five-seed headline backbones of Figure 1b with a two-hidden-layer MLP probe under the same validation protocol as the linear probe; MNIST with 1,000 images gives the same picture (the MLP probe is within 0.5 points of the linear probe for every head). With the original grid the MLP probe selected its lowest learning rate (10^{-3}) for the linear- and default-head backbones. Extending the grid down to 10^{-4} for the five headline heads (linear, default, ring-free, last ring, pairs; 5 seeds) changes every contrast against the default circuit by less than 0.1 points (VQC − linear -4.8 ± 0.7, every seed negative); one default-head probe still selects the new lowest rate. A stronger nonlinear decoder could still recover more; we claim only that the gap persists under these probes. Table 13 gives the effective rank of the backbone's loss gradient, and Table 14 the initialization control.

Tables 12 and 15 test whether the attenuation of Proposition 2 causes the cost. The first decoupling attempt passed each input's gradient only through its own factor, which discards the sign of the other factors' product as well as its magnitude, and training collapsed. The gradient-floor test keeps the forward products and each path's sign and floors only the magnitude, $\mathrm{sign}(P)\max(|P|, \tau)$ with the sign taken as $+1$ at $P = 0$ ($\tau = 0$ is the true gradient, tested to match it within numerical tolerance). At initialization, in training mode (a retrospective check), $\tau = 1$ changes the bottleneck gradient substantially (mean cosine with the true gradient 0.77–0.92 for the default and 0.63–0.73 for the last-ring supports across seeds; median norm ratios 1.3–1.6 and 1.8–4.1), so the floor acts from the start of training; a pre-run check in evaluation mode, with uncalibrated BatchNorm statistics, had understated this, and our expectation of a weak intervention rested on it. We predicted that $\tau = 1$ would recover at least 2 probe points on the default supports and 4 on the last-ring supports if attenuation drives the cost, and committed a training guard and a refutation criterion. On the default supports training stays within the guard and there is no observed mean recovery (-0.4 ± 2.0 at $\tau = 1$; approximate paired 95% interval $[-2.9, 2.1]$, which does not exclude the predicted 2 points), which meets that support's part of the refutation criterion; on the last-ring supports $\tau = 1$ lowers clean training accuracy by 11 points, beyond the guard, so that support is inconclusive and the joint criterion is not met; $\tau = 0.3$ has no reliable effect. We expected a partial recovery and found none. The attenuation hypothesis is therefore not supported: in this test, removing the attenuation of the other factors did not recover the representation. The test does not exclude attenuation acting otherwise, for instance inside trained circuits or through each input's own derivative; Proposition 2 remains a correct bound on gradient size, not a demonstrated mechanism.

Table 11. Linear vs nonlinear probe on the same frozen bottlenecks (CIFAR-10, 4,000 training images; MLP probe: standardization, then $8 \to 256 \to 256 \to 10$ with ReLU, probe learning rate and epoch selected on validation; mean $\pm$ SD over n seeds; last column paired). The MLP probe recovers at most a fraction of a point more than the linear probe for any head, and VQC $-$ linear is -4.8 ± 0.9 under the MLP probe vs -4.7 ± 0.9 under the linear probe.

Head the backbone was trained with	n	Linear probe	MLP probe	MLP $-$ linear
linear	5	74.5 ± 0.5	74.4 ± 0.5	0.0 ± 0.1
MLP-8	5	70.8 ± 0.5	70.8 ± 0.6	0.0 ± 0.6
VQC, no CNOT	5	73.6 ± 0.5	73.6 ± 0.4	0.0 ± 0.4
VQC, pairs	5	72.9 ± 0.6	72.9 ± 0.5	0.0 ± 0.4
VQC, pairs, last	5	71.2 ± 0.7	71.2 ± 0.5	0.0 ± 0.4
VQC, ring (default)	5	69.8 ± 1.0	69.6 ± 0.8	-0.2 ± 0.3
VQC, ring, last	5	60.1 ± 3.1	60.3 ± 2.7	$+0.2 \pm 0.9$
cos	5	73.3 ± 0.4	73.2 ± 0.6	-0.2 ± 0.3
cos-prod. (default supports)	5	68.8 ± 1.6	69.0 ± 1.5	$+0.2 \pm 0.2$
cos-prod. (last-ring supports)	5	54.8 ± 3.4	55.2 ± 3.3	$+0.4 \pm 0.4$

Table 12. Gradient-decoupled classical product heads: same forward products, but each input's gradient passes only through its own cosine factor (straight-through on the other factors). CIFAR-10, 4,000 images, 3 seeds; decoupled heads at their validation-selected learning rate (grid 10^{-3}–3×10^{-2}), coupled heads as in Table 8. Decoupling fails to fit the training set, so it does not isolate the effect of coupling.

		Clean-train acc.		Probe	
Products	lr	coupled	decoupled	coupled	decoupled
default supports	0.003	84.0 ± 1.7	55.7 ± 3.7	67.8 ± 1.1	57.7 ± 1.5
last-ring supports	0.003	56.4 ± 5.2	42.4 ± 5.3	54.2 ± 3.1	48.9 ± 4.7

G. A Second Family of CNOT Layouts

Table 16 tests product order out of sample. The six layouts were chosen by their zero-rotation order alone, from CNOT patterns not used anywhere else in the paper, and their predicted probe accuracies and the pass criteria (Spearman correlation at most -0.8; at least four of six within 4 points) were recorded before any run. Every layout selected the head learning rate 10^{-2}, interior to its grid. Both criteria pass, but the order ranking is not fully preserved: two pairs reverse it in every seed (brickwork $-$ distance-2 ring after the last layer $+4.0 \pm 0.7$; star-shaped ring after the last layer $-$ open chain $+1.7 \pm 1.3$). The star-shaped ring gives one readout order 8 and the other seven order 1, and probes 2.4 points above the prediction from its mean order of 1.875; a single high-order readout may cost less than the mean suggests, a hypothesis we have not tested. Brickwork, which alternates its pairs between layers, probes 5.1 points above its prediction. Its supports are nearly uniform (every input in 3 or 4 of them), but so are those of the distance-2 ring after the last layer (2 to 4), which lands on its prediction, so support uniformity does not explain the miss. What does separate the two misses is how far training moves each circuit from its zero-rotation model (post hoc, Table 16): brickwork's trained readouts are the least well reproduced by its products (held-out R^2 0.51, against 0.70–0.89 for the others; trained accuracy 70.2 vs 45.0 with the remapped products), it has the largest trained R_y rotations (mean $|\theta|$ 0.44), and the star-shaped ring has the next-lowest R^2 (0.70). Across the six layouts the prediction residual and R^2 have rank correlation -0.94. Order describes a circuit at its zero-rotation start, and layouts whose trained rotations carry them furthest from it escape part of the predicted cost; with six layouts this is an association, not a mechanism. Measured directly, brickwork's effective order at its trained rotations is 2.6 rather than 3.5 (Section N). The gradient effective rank of Table 13 (star 7.1, chain 5.7, star-shaped ring 6.5, distance-2 ring 4.6, brickwork 4.7, chain after the last layer 3.5) ranks the six layouts exactly as their probe accuracy does, including both reversals, but captures little of brickwork's advantage: its rank exceeds the distance-2 ring's by only 0.1 ± 0.6, against a 4-point probe gap. A two-hidden-layer MLP probe gives the same pattern (brickwork 70.2, open chain after the last layer 58.8).

H. Product Order under a Second Encoding

Every R_y angle encoding satisfies Equation (1), so a change of angle map keeps each circuit's product order but changes the factors. Under $a_j = \frac{\pi}{2} \tanh z_j + \frac{\pi}{2}$ (the monotone encoding of Table 10) each factor is $-\sin(\frac{\pi}{2} \tanh z_j)$: it vanishes at $z_j = 0$, where $\cos(\pi \tanh z_j)$ is 1, so attenuation is strongest exactly where the default encoding has none. We trained six layouts of orders 1 to 5.25 with it, four from the ring family and two from the second family, committing predictions and pass criteria to the repository before the new runs; the default ring's three existing seeds supply the anchor (Table 17). All

Table 13. Effective rank (of 8) of the class signal the backbone receives: $\exp H(p)$, with p the normalized eigenvalues of $\frac{1}{N}\sum_i g_i g_i^\top$, $g_i = \partial \mathcal{L}_i / \partial z$ (cross-entropy) on the 2,000 validation embeddings (CIFAR-10, 4,000 training images; 5 seeds). Final: the trained run at its validation-selected lr. Init: a freshly initialized model of the same kind built with the run's seed. Δ: paired difference to the default circuit.

		Init		Final	
Head	Order	rank	Δ	rank	Δ
linear	—	5.4 ± 0.5	$+2.6 \pm 0.5$	7.3 ± 0.2	$+2.8 \pm 0.7$
VQC, no CNOT	1	4.0 ± 0.6	$+1.2 \pm 0.8$	7.2 ± 0.2	$+2.7 \pm 0.8$
VQC, ring (default)	2.5	2.8 ± 0.4	—	4.5 ± 0.6	—
VQC, ring, last	5.25	1.7 ± 0.3	-1.1 ± 0.4	3.7 ± 0.6	-0.8 ± 0.6

Table 14. Initialization control: probe accuracy (%) with the variational rotations initialized in $\mathcal{U}(-0.1, 0.1)$ (near the identity, the protocol default) or $\mathcal{U}(-\pi, \pi)$ (CIFAR-10, 4,000 training images; seeds 0–2; head lr selected on validation from $\{3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$ at the large initialization). The last row is the paired difference between the two circuits.

Circuit	$\mathcal{U}(-0.1, 0.1)$	$\mathcal{U}(-\pi, \pi)$	large − small	selected lr
VQC, ring (default)	69.9 ± 1.1	69.0 ± 1.8	-0.9 ± 1.6	0.01 / 0.01
VQC, no CNOT	73.7 ± 0.6	73.6 ± 0.4	-0.1 ± 0.2	0.01 / 0.01
no CNOT − default	$+3.8 \pm 1.6$	$+4.6 \pm 1.7$		

criteria are met: the rank correlation with order is -0.94 under both probes, each of the five predicted layouts lands within 4 points, and removing the CNOTs recovers 3.0 ± 0.5 points over the default ring, positive for every seed. The one rank reversal, chain-last (order 4.5) below ring-last (5.25), was named as a risk in advance and appears in the same direction under $\pi \tanh$. The predicted slope, scaled down from the ring family's by the measured attenuation factor, matches the observed one (-2.41); post hoc, the same layouts under $\pi \tanh$ give a steeper slope, consistent with the attenuation account of Proposition 2 but resting on six layouts; the direct test of that account on the $\pi \tanh$ product heads (Table 15) finds no recovery on the default supports and is inconclusive on the last-ring supports. Section I turns to an encoding for which Equation (1) does not hold.

I. An Encoding Whose Zero-Rotation Readouts Vanish

With the phase encoding, H then $R_z(\pi \tanh z_j)$, each qubit's Bloch vector is $(\cos a_j, \sin a_j, 0)$, so every readout is zero at zero rotations and the product order of Equation (1) is undefined. The effective order of Section N still applies, with Z instead of Y strings invisible (the expansion and the simulator are tested against an independent PennyLane circuit). A last-layer circuit's readouts factorize over its supports at any rotations, so its effective order equals its R_y order whenever the readout is nonzero. At the initialization distribution the effective order therefore differs from the R_y order only on the three layouts with CNOTs after every layer, and there probe accuracy follows the R_y order: the star, at initialization order 4.5 but R_y order 1, probes like the no-CNOT head (Table 18). We committed both orders as competing predictors before the runs, expecting the initialization order to predict moderately and the R_y order to fail; the R_y order passed. The initialization orders we committed were computed with a numerical cutoff that zeroed tiny but genuine last-layer readouts (first column) and failed as committed; the correct values (second column) meet the criterion only in retrospect. The no-CNOT head landed just below its predicted range (71.8 against $[72, 75]$). Post hoc, the trained rotations agree: on average over readouts about 99% of the weight is invisible at initialization, after training it is near zero for every layout except the chain and the default ring (0.30 and 0.48), and the trained effective order ranks the layouts as probe accuracy does. The two highest-order heads fit their training data poorly under this encoding (54–57%), vary widely across seeds, and probe about 7 points below their R_y-order predictions.

J. Width: 12 Qubits

Table 19 repeats the headline comparison with a 12-dimensional bottleneck and 12 qubits, one per coordinate, under the same protocol; Equation (1) holds at this width (tested), with product orders 1, 3.92 and 7.33 for the circuits without CNOTs, with the default rings and with a ring after the last layer only. We committed the predictions before the runs: the ordering of the three circuits, a default circuit at least 2 points below the ring-free one and below the linear head, and, as an order-only extrapolation, gaps to the ring-free head of 9.4 and 20.4 points from the 8-qubit ring-family slope, which we expected to

Table 15. Gradient-floor test of the attenuation hypothesis (CIFAR-10, 4,000 images, small CNN, 5 seeds, head learning rate 10^{-2} selected for every head). Classical product heads keep the forward products; in the backward pass each path's derivative, the product P of the other factors in the readout's support, is replaced by $\mathrm{sign}(P)\max(|P|,\tau)$ (sign $+1$ at $P=0$): $\tau=0$ is the true gradient, $\tau=1$ removes all magnitude attenuation but keeps each path's sign. Predictions, a training guard (clean training accuracy within 10 points of $\tau=0$) and a refutation criterion were committed before the runs. Δ: paired probe difference to $\tau=0$. Train: clean training accuracy (%).

Supports	τ	Probe	Δ	MLP probe	Train
default (order 2.5)	0	68.8 ± 1.6	—	69.0 ± 1.5	86.0
	0.3	68.3 ± 2.0	-0.5 ± 2.5	68.8 ± 2.0	85.2
	1	68.4 ± 2.2	-0.4 ± 2.0	68.2 ± 2.3	84.1
last ring (order 5.25)	0	54.8 ± 3.4	—	55.2 ± 3.3	56.1
	0.3	56.6 ± 6.2	$+1.7 \pm 7.3$	57.3 ± 5.7	58.8
	1	47.4 ± 3.1	-7.4 ± 4.7	48.2 ± 3.1	44.7

Table 16. Out-of-sample test of product order on a second family of CNOT layouts (CIFAR-10, 4,000 images, small CNN, protocol of Section 3). The layouts were chosen by their zero-rotation product order alone, and their probe accuracy was predicted, before training, from a linear fit over the seven ring-family VQC circuits, seeds 0–2 (probe $= 76.7 - 3.22\times$ order). Spearman correlation between order and mean probe accuracy over the six layouts: -0.89; 5 of 6 within 4 points of the prediction. R^2: held-out fraction of the trained readouts' variance recovered by an affine map of the same circuit's zero-rotation readouts (Section B); its rank correlation with the prediction residual (observed − predicted, post hoc) is -0.94.

Layout	CNOTs	Order	Predicted	Probe	Obs. − pred.	R^2	n
star	$\mathrm{CNOT}(0,q)$, every layer	1	73.5	73.1 ± 1.2	-0.4	0.82	5
chain	$\mathrm{CNOT}(q,q{+}1)$, $q<7$, every layer	1.5	71.9	71.4 ± 1.1	-0.5	0.84	5
star-in, last	$\mathrm{CNOT}(q,0)$, last layer	1.875	70.7	73.0 ± 0.5	+2.4	0.70	5
ladder-2, last	$\mathrm{CNOT}(q,q{+}2)$, last layer	3	67.0	66.6 ± 0.5	-0.5	0.81	5
brickwork	alternating pairs, every layer	3.5	65.4	70.5 ± 0.6	+5.1	0.51	5
chain, last	open chain, last layer	4.5	62.2	58.2 ± 3.1	-4.0	0.89	5

overshoot. The ordering and both secondary criteria hold in every seed, under both probes. The gaps, however, barely change with width: the default circuit sits 4.0 points below the ring-free one (3.8 at 8 qubits) and the last-ring circuit 12.3 points below it (13.5), although their orders rose from 2.5 and 5.25. Product order therefore ranks these three circuits at both widths, but the cost per unit of order is not constant across widths, and the 8-qubit slope does not transfer.

K. Bottleneck Saturation

Tables 20 and 21 test where the cost arises, in three pre-registered steps. First, on the trained backbones of 38 configurations (both layout families, the default and two alternative encodings, both widths and the classical product heads), we measured the fraction of saturated bottleneck coordinates, the effective dimensionality of the bottleneck and that of its class means, and committed in advance that a candidate must correlate with probe accuracy at $\rho \le -0.7$ (saturation) or $\rho \ge 0.7$ (dimensionality) over the VQC and over the classical cells. All three pass. We had expected saturation to track the probe poorly across encodings, and it did not. A secondary criterion also passes: within the 13 8-qubit layouts, the part of effective dimensionality not explained by order tracks the probe's residual from the committed ring-family fit of probe on order ($\rho = 0.87$; criterion ≥ 0.5). Saturation and dimensionality are themselves tightly coupled ($\rho = -0.91$ across cells), so the diagnostics alone cannot order them.

Second, in a discovery cohort (seeds 0–4), we added one penalty to the loss of the classical product heads, either against saturation or against correlated bottleneck coordinates, with the weight selected on validation accuracy. Both pass their manipulation checks (last-ring saturation ≤ 0.42 under the first: 0.03; an effective-dimensionality increase of at least 0.05 under the second: 0.28 to 0.93), and both improve fit. On the default products the anti-saturation penalty meets every criterion and decorrelation does not; on the last-ring products both raise the probe substantially (Table 21). We had expected the anti-saturation penalty to fail; the decorrelation pattern, larger on the last-ring than on the default products, was as expected, its default gain below the criterion. The MLP probe agrees (anti-saturation $+0.5$, $+2.0$, $+12.2$; decorrelation -0.4, $+0.0$, $+8.0$ on the order-1, 2.5 and 5.25 heads). At the non-selected weights (3 seeds), anti-saturation gives $+0.6$, $+2.9$ and $+10.7$ points, and decorrelation -3.0, -0.3 and $+3.7$.

Table 17. Product order under the monotone encoding $a_j = \frac{\pi}{2}\tanh z_j + \frac{\pi}{2}$ (CIFAR-10, 4,000 images, small CNN, protocol of Section 3; every layout selected the head learning rate 10^{-2}). Each zero-rotation factor becomes $\cos a_j = -\sin(\frac{\pi}{2}\tanh z_j)$, which vanishes at $z_j = 0$ where $\cos(\pi \tanh z_j)$ is 1; product orders are unchanged. Predictions were committed before the new runs: the default ring's observed probe (3 existing seeds) plus a slope of -2.45 per unit of order, the ring-family slope scaled by the ratio of $\ln \mathbb{E}[\cos^2 a_j]$ under the two encodings measured on the trained default ring (0.71 vs 0.63, Section D). Spearman correlation between order and mean probe accuracy: -0.94 (criterion ≤ -0.8; MLP probe -0.94); 6 of 6 within 4 points (including the anchor); no CNOT − default $+3.0 \pm 0.5$ (criterion ≥ 2). Post hoc, the probe-on-order slope is -2.41 here and -3.59 for the same layouts under $\pi \tanh$ (last column), a difference of $+1.18$ (95% percentile interval over all 5^5 resamples of the 5 shared seeds, $[+0.82, +1.56]$).

Layout	Order	Predicted	Probe	Obs. − pred.	MLP probe	Probe, $\pi \tanh$	n
no CNOT	1	73.9	73.2 ± 0.6	-0.7	73.2 ± 0.7	73.6 ± 0.5	5
pairs, last	1.5	72.6	71.8 ± 0.8	-0.8	71.5 ± 1.1	71.2 ± 0.7	5
ring (default)	2.5	70.2	70.2 ± 0.7	+0.0	70.0 ± 0.6	69.8 ± 1.0	5
ladder-2, last	3	69.0	67.7 ± 1.1	-1.3	67.7 ± 0.7	66.6 ± 0.5	5
chain, last	4.5	65.3	62.8 ± 2.7	-2.5	63.2 ± 2.5	58.2 ± 3.1	5
ring, last	5.25	63.5	64.3 ± 2.3	+0.9	64.9 ± 2.2	60.1 ± 3.1	5

Table 18. The phase encoding, H then $R_z(\pi \tanh z_j)$ (CIFAR-10, 4,000 images, small CNN, protocol of Section 3; every layout selected the head learning rate 10^{-2}). Every zero-rotation readout vanishes, so Equation (1) does not apply. Two competing predictions were committed before the runs: the effective order (Section N) at the initialization distribution $\theta \sim \mathcal{U}(-0.1, 0.1)$, and the R_y product order. As committed (first column), the initialization order correlated -0.23 with the R_y order and $+0.09$ with probe accuracy (criterion ≤ -0.8; failed; MLP probe $+0.09$), and its point predictions put 2 of 6 layouts within 4 points; the R_y order correlated -0.99 (passed; MLP probe -0.99) with 4 of 6 within 4 points. After the runs we found that the committed values came from code that set a readout's order to 0 when its total visible weight was below 10^{-12}, which happens for the last-layer layouts at small rotations; with the definition applied (second column, a retrospective correction) the initialization order equals the R_y order for those layouts and correlates -0.83 with probe accuracy. Post hoc: the effective order at the trained rotations, and the share of weight on strings invisible to the encoded state (about 0.99 at initialization, averaged over readouts), with a rank correlation of -1.00 between trained order and probe accuracy. Train: clean training accuracy (%).

	Predictors			Trained (post hoc)					
	Init. order								
Layout	committed	corrected	R_y order	Order	Invisible	Probe	MLP probe	Train	n
no CNOT	1.00	1.00	1	1.00	0.00	71.8 ± 1.0	71.7 ± 1.1	92.6	5
star	4.47	4.47	1	1.09	0.00	71.5 ± 1.5	71.6 ± 1.3	91.6	5
chain	2.98	2.98	1.5	2.51	0.30	68.7 ± 1.7	68.8 ± 1.6	85.3	5
ring (default)	4.44	4.44	2.5	3.64	0.48	67.5 ± 0.8	67.6 ± 0.8	81.9	5
chain, last	1.64	4.50	4.5	4.50	0.00	55.3 ± 9.6	56.1 ± 9.0	57.1	5
ring, last	1.51	5.25	5.25	5.25	0.00	52.9 ± 11.1	53.6 ± 11.2	54.2	5

Third, the same anti-saturation penalty on the 8-qubit circuits passes its manipulation check (saturation 0.30 to 0.002 and 0.49 to 0.005, against thresholds of 0.18 and 0.30), improves fit, and raises probe accuracy on the default and last-ring circuits, positive in every seed under the linear probe, as predicted, with a small gain on the ring-free circuit ($+0.6 \pm 0.8$, within the specificity criterion); the MLP probe gives $+1.6$, $+7.7$ and $+0.4$ against its extended-grid baselines (one negative default seed). The gains recover about 44% and 59% of the two circuits' gaps to the unregularized ring-free head; the rest remains unexplained, and the refutation condition (gains below $+1$ on both) is not met.

Finally, a replication cohort (seeds 5–9). The training guard we committed, clean training accuracy within 10 points of the unregularized head, was meant to exclude lost fit but as written also excluded the large gains in fit on the last-ring heads (13.6 to 23 points), which left those results inconclusive. We therefore re-ran both arms of the last-ring comparison on fresh seeds, with the penalty fixed at 10 and a guard committed as no loss of fit. The replication meets every criterion: $+6.0 \pm 4.1$ points for the last-ring circuit and $+9.9 \pm 5.2$ for its products, positive in every seed, with saturation down to 0.01 and 0.04 and unregularized baselines of 60.0 and 57.2 (60.1 and 54.8 on seeds 0–4); the MLP probe gives $+6.1$ and $+9.4$. The penalty acts during training only, so the heads keep the same forward architecture. Under criteria as committed, the results support a causal role for saturation on both supports and both head families; the penalty also changes fit and effective dimensionality, so saturation need not be the only route.

Table 19. Width replication (CIFAR-10, 4,000 images, small CNN, protocol of Section 3, 5 seeds): the headline heads with an 8-dimensional bottleneck and 8 qubits, and with a 12-dimensional bottleneck and 12 qubits. Product orders follow Equation (1) with q the number of qubits. Predictions for the 12-qubit runs were committed before training: the ordering no CNOT > default > last ring (met), and a paired mean default − no CNOT ≤ -2 and default − linear < 0, each negative in at least 4 of 5 seeds (met: -4.0 ± 1.0 and -4.9 ± 1.0, every seed negative). Extrapolating the 8-qubit ring-family slope (3.22 points per unit of order) predicted gaps of -9.4 and -20.4 to the no-CNOT head; the observed gaps are much smaller and close to those at 8 qubits. Δ: paired probe difference to the no-CNOT head. Selected head learning rates at 12 qubits: 3×10^{-2} (linear, after one edge step) and 10^{-2} (VQC). Train: clean training accuracy at 12 qubits (%).

	8 qubits			12 qubits				
Head	Order	Probe	Δ	Order	Probe	Δ	MLP probe	Train
linear	—	74.5 ± 0.5	—	—	74.7 ± 0.7	—	74.5 ± 0.7	98.7
VQC, no CNOT	1	73.6 ± 0.5	—	1	73.8 ± 0.5	—	73.5 ± 0.9	97.0
VQC, ring (default)	2.50	69.8 ± 1.0	-3.8 ± 1.2	3.92	69.8 ± 0.8	-4.0 ± 1.0	69.7 ± 0.9	86.3
VQC, ring, last	5.25	60.1 ± 3.1	-13.5 ± 3.1	7.33	61.5 ± 6.1	-12.3 ± 6.2	61.6 ± 6.5	66.9

Table 20. Bottleneck geometry of the trained backbones (CIFAR-10, 4,000 images, small CNN; test embeddings z of every probed run). S: fraction of coordinates with $|\tanh z| > 0.9$; PRn: participation ratio of the correlation matrix of z, divided by the width; CPRn: participation ratio of the between-class covariance of standardized z, divided by $\min(q, 9)$. Entries: Spearman correlation between the cell means and the cells' mean probe accuracy. Criteria ($\rho \leq -0.7$ for S, $\rho \geq 0.7$ for PRn and CPRn, separately over the VQC and the classical cells) were committed before the computation.

Cells	n	S	PRn	CPRn
VQC, 8 qubits ($\pi \tanh$)	13	-0.97	0.99	0.97
VQC, half encoding	6	-1.00	0.94	0.94
VQC, phase encoding	6	-1.00	0.94	0.94
VQC, 12 qubits	3	-1.00	1.00	1.00
classical product heads	10	-0.98	0.96	0.95
all VQC cells	28	-0.91	0.86	0.92

L. The Signal Budget

Proof of Proposition 1. Write $c_j = \cos a_j$ and $t_j = |\tanh z_j|$. Independence gives $\mathbb{E}|r_k| = \prod_{j \in S_k} \mathbb{E}|c_j| \geq a$, so the largest $\mathbb{E}|c_j|$ is at least $a^{1/m}$. On the middle zone $\delta < t_j < 1 - \delta$ we have $|c_j| = |\cos \pi t_j| \leq \cos \pi\delta$, and $|c_j| \leq 1$ everywhere, so $\mathbb{E}|c_j| \leq 1 - (1 - \cos \pi\delta) \Pr(\delta < t_j < 1 - \delta)$. Rearranging gives the bound. With identically distributed factors every $\mathbb{E}|c_j|$ equals the largest. □

Without independence. Since $|c_i| \leq 1$, $|r_k| \leq |c_j|$ for every $j \in S_k$, so $\mathbb{E}|c_j| \geq a$ and the bound holds for every input with a in place of $a^{1/m}$; this version needs no assumption on the joint distribution but loses the dependence on order. Some assumption is needed for that dependence: if all inputs of a support are equal, $r_k = c^m$ and $\mathbb{E}|c|^m \geq a$ allows a middle-zone mass up to $(1 - a)/(1 - \cos^m \pi\delta)$, which tends to $1 - a$ rather than to 0 as m grows. (Generalized Hölder, $\mathbb{E} \prod_j |c_j| \leq \prod_j (\mathbb{E}|c_j|^m)^{1/m}$, gives this bound for some j under any dependence.) The stated order dependence therefore assumes independence; its robustness to dependence is not established. On the trained bottlenecks, the measured $\mathbb{E}[\prod_i \cos^2 a_i]$ matches the independence product to within 0.04 in the moments measured (Section D), which is consistent with, but does not establish, near-independence.

The proposition says the middle zone must empty as order grows; it does not say at which end the coordinates gather. At $z \approx 0$ a coordinate gives $c = +1$, and at saturation $c = -1$ with $\operatorname{sech}^2 z \approx 0$. Near either end a coordinate's factor is locally insensitive to it, so graded variation in a high-order support has little effect on the readout. Averaged over a support, the bound gives $\frac{1}{m} \sum_j \Pr(\text{middle}) \leq -\log a / \big(m(1 - \cos \pi\delta)\big)$ under independence, so the mean middle-zone mass vanishes as m grows, though a single coordinate need not leave it. The same argument holds for any encoding whose factors are bounded by 1 and are flat only at the ends, including $a_j = z_j$, with the zones defined around the extrema of $\cos z$ at multiples of π.

Premise and tightness. The proposition is conditional on $\mathbb{E}|r_k| \geq a$ for the zero-rotation products r_k, with a fixed as order grows, and two things could defeat its use here. First, a learned linear decoder could let readouts shrink and scale

Table 21. Interventions on bottleneck saturation (CIFAR-10, 4,000 images, small CNN, head learning rate 10^{-2}, 5 seeds). Each head is trained with a penalty on its bottleneck added to the loss: anti-saturation, $\lambda\,\mathrm{mean}(\mathrm{relu}(\tanh^2 z - 0.81))$, nonzero only where $|\tanh z| > 0.9$; decorrelation, λ times the mean squared off-diagonal entry of the batch correlation matrix of z. λ was selected on validation accuracy from $\{3, 10\}$ (anti-saturation) and $\{1, 3\}$ (decorrelation); every product head selected 10 and 1. Δ: paired probe difference to the unregularized head (linear probe). Manipulation checks, a training guard and the causal criteria (mean gains of at least $+3$ and $+1.5$ probe points on the order-5.25 and order-2.5 heads, positive in at least 4 of 5 seeds, with less than half the order-5.25 gain on the order-1 head) were committed before the runs. Anti-saturation meets the gain, sign and specificity criteria on both head families and decorrelation does not. The last-ring comparison was replicated on fresh seeds 5–9 (both arms re-run, penalty fixed at 10), where it meets every criterion; see Section K for why.

Head (order)	Probe	Δ anti-sat.	Δ decorr.	S	Clean train	λ
Classical product heads						
cos (1)	73.3 ± 0.4	$+0.2 \pm 1.1$	-0.3 ± 0.4	$0.05 \to 0.02$	$95.5 \to 96.6$	10
default supports (2.5)	68.8 ± 1.6	$+2.2 \pm 1.7$	$+0.3 \pm 1.9$	$0.41 \to 0.03$	$86.0 \to 91.7$	10
last-ring supports (5.25)	54.8 ± 3.4	$+12.6 \pm 4.9$	$+7.8 \pm 4.1$	$0.60 \to 0.03$	$56.1 \to 78.8$	10
8-qubit VQC heads						
no CNOT (1)	73.6 ± 0.5	$+0.6 \pm 0.8$	—	$0.01 \to 0.00$	$96.6 \to 97.7$	3
ring, default (2.5)	69.8 ± 1.0	$+1.7 \pm 1.3$	—	$0.30 \to 0.00$	$89.2 \to 93.6$	10
ring, last (5.25)	60.1 ± 3.1	$+7.9 \pm 3.0$	—	$0.49 \to 0.00$	$65.6 \to 79.3$	10
Replication on seeds 5–9, both arms (guard: no loss of fit)						
VQC, ring, last (5.25)	60.0 ± 4.4	$+6.0 \pm 4.1$	—	$0.46 \to 0.01$	$64.6 \to 76.7$	10
last-ring products (5.25)	57.2 ± 5.2	$+9.9 \pm 5.2$	—	$0.62 \to 0.04$	$60.2 \to 78.2$	10

them back up: factors $y/2$ and $\frac{1}{2}$ give $r_k = 2^{-m}y$, which a weight of 2^m restores. A pre-registered measurement on the 61 trained 8-qubit runs behind Table 20 (13 layouts, orders 1–5.25, test split) finds no such compensation in the circuits' actual outputs, which are what the decoder sees. Their mean magnitude $\mathbb{E}|\langle Z_k\rangle|$ is 0.66 without CNOTs, 0.51 for the default ring, 0.52 for the last ring and at least 0.39 in every layout (Spearman -0.88 with order), and the mean norm of the decoder column attached to each readout shrinks rather than grows (5.72 to 4.05; Spearman -0.54), so the logit scale each readout contributes falls with order (4.02 to 2.31, Spearman -0.91, as for probe accuracy over the same layouts). Second, actual outputs are not the proposition's object: trained rotations can make them large while the zero-rotation products vanish ($R_y(-\pi/2)$ on every qubit before the last ring maps the encoded state at $\tanh z_j = \frac{1}{2}$ to $|0\rangle^{\otimes 8}$, so every output is 1 and every product 0). After the pre-registered analysis we therefore computed the proposition's own quantity on the same bottlenecks, the per-sample products $r_k = \prod_{j\in S_k} \cos a_j$. Averaged over readouts, $\mathbb{E}|r_k|$ is 0.61 without CNOTs, 0.47 for the default ring, 0.44 for the last ring and at least 0.29 in every layout, and its layout means are within 0.04 of the product of marginals $\prod_{j\in S_k} \mathbb{E}|\cos a_j|$ (individual runs within 0.07), consistent with near-independence at this moment. The bound is nonetheless loose at these orders. For readouts of order at least 4, $\mathbb{E}|r_k|$ is 0.16–0.24 (layout means of the smallest such readout; 0.02 for the order-8 readout of the star-into-last layout), and the bound with that a falls below 1 only for δ above 0.20–0.29 (layout means of each run's largest per-readout threshold; at most 0.31 in any run). It constrains roughly the centre $0.3 < |\tanh z| < 0.7$, not the whole graded zone, and gives the direction of the depletion in Table 20, not its size. All of these are measured at the end of training; how they evolve during training is not measured. Nor does depletion by itself imply lost task information: the bound allows one graded, informative coordinate per support, and the link between depletion and probe accuracy rests on the measurements of Table 20 and on the escape-only intervention, not on the proposition.

Proposition 3 (Flat ends)**.** *Let* $t = |\tanh z|$ *and* $g(z) = \left|\frac{d}{dz}\cos(\pi \tanh z)\right|$. *Then* $g(z) \le \pi^2 t$, $g(z) \le 2\pi^2(1-t)^2 \le 8\pi^2 e^{-4|z|}$, *and* $g(z) > \pi\delta \sin \pi\delta$ *if* $\delta < t < 1-\delta$. *Under the assumptions of Proposition 1 with identically distributed factors,* $\Pr\left(|\partial r_k/\partial z_j| > \pi^2\delta\right) \le (1 - a^{1/m})/(1 - \cos\pi\delta)$ *for every* $j \in S_k$.

Proof of Proposition 3. $\cos(\pi \tanh z)$ is even in z, so take $z \ge 0$; then $g(z) = \pi(1-t^2)\sin \pi t$. Near zero, $\sin \pi t \le \pi t$ and $1 - t^2 \le 1$, so $g \le \pi^2 t$. Near saturation, $\sin \pi t = \sin \pi(1-t) \le \pi(1-t)$ and $1 - t^2 = (1-t)(1+t) \le 2(1-t)$, so $g \le 2\pi^2(1-t)^2$; and $1 - t = 2/(e^{2z}+1) \le 2e^{-2z}$ gives $g \le 8\pi^2 e^{-4z}$. On $\delta < t < 1-\delta$, concavity and symmetry of $\sin \pi t$ on $[0,1]$ give $\sin \pi t \ge \sin \pi\delta$, and $1 - t^2 > 1 - (1-\delta)^2 = \delta(2-\delta) > \delta$. For the last claim, $t \le \delta$ gives $g \le \pi^2\delta$ and $t \ge 1-\delta$ gives $g \le 2\pi^2\delta^2 \le \pi^2\delta$ since $\delta < \frac{1}{2}$, so $g(z_j) > \pi^2\delta$ only on the middle zone. By Equation (3) and $|\cos a_i| \le 1$, $|\partial r_k/\partial z_j| \le g(z_j)$, and Proposition 1 bounds the middle-zone probability. □

Two further facts: $1 - t \ge e^{-2z}$ for $z \ge 0$ gives $\mathrm{sech}^2 z = (1-t)(1+t) \ge e^{-2z}$, and $1 - \cos x \le x^2/2$ with $\cos \pi t = -\cos \pi(1-t)$ gives $|\cos a_j + 1| \le \frac{\pi^2}{2}(1-t)^2 \le 2\pi^2 e^{-4z}$. The flatness at saturation thus has two sources, $\mathrm{sech}^2 z$

and $\sin a_j$, each decaying as $e^{-2|z|}$. The escape-only estimator of Section 5 removes only the first, so for a cosine-product readout a saturated coordinate's inward gradient still decays as $e^{-2|z|}$, the rate of a plain $\tanh$ bottleneck, rather than not at all. This holds for the zero-rotation products only: a trained circuit readout can contain a component $\beta_j \sin a_j$, whose surrogate inward gradient tends to $\pi|\beta_j|$ rather than to zero. With the non-saturating encoding $a_j = z_j$, $g(z) = |\sin z|$ has no flat tail, only isolated zeros at multiples of π. The bounds concern the readout's local sensitivity, not the loss. A coordinate that the budget pushes to either end carries little gradient through its own readouts, but the bounds do not show that gradient descent cannot move it there or out again.

Where the trained coordinates sit. On the test embeddings of the archived heads (CIFAR-10, 4,000 images, seeds 0–4), the middle-zone fraction M and the mean factor magnitude $\mathbb{E}|\cos a_j|$ are 0.84 and 0.61 without CNOTs, 0.63 and 0.73 for the default circuit, and 0.46 and 0.82 for the last-ring circuit; for the default and last-ring products, 0.53 and 0.78, and 0.35 and 0.86. The near-zero fraction falls with order (0.15, 0.08, 0.05), so the mass leaves the middle for saturation.

A toy model. A 32–64–8 MLP on a ten-class Gaussian mixture, trained through classical product heads on the paper's supports (5 seeds), reproduces the falling middle zone and a probe cost that generally rises with order (with reversals at orders 3 and 4), but a smaller one (3.7 points from order 1 to 5.25) and with coordinates pinned near $z \approx 0$ rather than at saturation. Tripling the backbone's initial output scale moves the pinning to saturation without enlarging the default support's cost against order 1 (-2.4 ± 1.0 at the original scale, -2.3 ± 0.7 at three times it). The toy is consistent with an order-associated cost without predominant saturation; in the CNNs, which are saturated by epoch 5, the interventions of Section M support an additional role for saturation.

M. Training Dynamics and Interventions on the Saturation Trap

Dynamics and intervention runs: CIFAR-10, 4,000 images, small CNN, learning rate 10^{-2} (the selected rate of the unregularized heads), 80 epochs. Predictions were committed before each batch of runs.

Dynamics. We retrained the ring-free, default and last-ring circuits (seeds 0–2) with a checkpoint every 5 epochs. These runs reproduce the archived per-seed probes exactly. At each checkpoint we fit the common linear probe and measured S and M (Table 22). Every head saturates early, and the heads then differ in how far they desaturate. We had pre-registered two timing hypotheses that assumed saturation would rise during training (saturation's half-way rise preceding the probe gap's, or the middle zone's drop preceding saturation's). Saturation is past its half-way point at the first checkpoint (epoch 5) in every run, while the probe gaps reach half their final size at epoch 5 or 10. The first hypothesis, stated as "no later than", is therefore met as written in all three seeds, but only because saturation is already high at the first checkpoint, so it does not establish that saturation comes first; the second cannot be scored at this resolution and is inconclusive. We then pre-registered a follow-up with a checkpoint after each of epochs 1–10 (same runs, which reproduce the 5-epoch checkpoints): the last-ring head's saturation should exceed the ring-free head's by at least 0.10 no later than the epoch at which the probe gap first reaches half its maximum, in at least 2 of 3 seeds. It holds in all three: the saturation gap is already 0.18–0.32 after epoch 1, while the probe gap first reaches half its maximum at epochs 8, 2 and 9. Seed-mean saturation peaks after one or two epochs, at 0.55 (ring-free), 0.78 (default) and 0.82 (last ring). The probe gap peaks at epoch 15 (default) and 35 (last ring) and then partly closes as S slowly falls, consistent with an incomplete escape under this training budget; this agrees with the doubled-budget results of Section 5.5.

Table 22. Training dynamics (seed means, 3 seeds). S: saturated fraction; gap: probe accuracy minus the ring-free circuit's at the same epoch and seed; M: middle-zone fraction (ring-free in parentheses).

	S			gap		M
Epoch	no CNOT	ring	last	ring	last	last (no CNOT)
5	0.44	0.67	0.77	−3.1	−8.4	0.21 (0.51)
10	0.23	0.56	0.68	−7.9	−14.3	0.30 (0.69)
20	0.12	0.42	0.59	−7.9	−18.8	0.38 (0.80)
40	0.02	0.34	0.48	−4.9	−17.2	0.47 (0.86)
80	0.02	0.30	0.46	−3.6	−12.7	0.48 (0.85)

Straight-through estimators. A plain straight-through estimator replaces $\mathrm{d}\tanh z/\mathrm{d}z$ by 1 in the backward pass and leaves the forward pass unchanged. On the classical product heads it raised the last-ring probe by 5.7 points and lowered the default products' by 1.0. On every circuit it diverged (head-mean validation accuracy 18–25%, backbone gradient norm up to 2.4×10^4): a sinusoidal component $\beta_j \sin a_j$ of a circuit readout keeps a gradient of order $\pi\beta_j$ at saturation while its forward value is flat, which can drive z_j further out with only an exponentially small change in the loss; we take this as the likely cause. Under the committed training guard these cells are inconclusive. After seeing this we pre-registered the escape-only estimator, which uses $\mathrm{d}\tanh z/\mathrm{d}z = 1$ only when the descent step moves z toward 0 and $\mathrm{sech}^2 z$ otherwise, per sample and coordinate. Its manipulation check (saturation down by at least 30%) and training guard (clean training accuracy no more than 10 points below baseline) pass in every cell; clean training accuracy rises. The committed hypothesis required paired gains of at least $+1.5$ (default) and $+3$ (last ring), each positive in at least four of five seeds, and a ring-free gain below half the last-ring gain. The observed $+1.48$, $+6.83$ and $+0.89$ (5 of 5 seeds positive in each) miss the first threshold by 0.02; the refutation condition (gains below $+1$ on both circuits) is not met. We had expected weaker results (a default gain of 0 to $+1.5$ and saturation falling by less than 30% on the default circuit). Because the default gain missed by so little, we pre-registered a replication on fresh seeds 5–9 with the same design and criteria (the unregularized last-ring arm is the retest arm on the same seeds); the seeds 0–4 outcome stands as scored. The replication meets every criterion: paired gains of $+3.30 \pm 3.02$ on the default circuit ($68.3 \to 71.6$), $+5.56 \pm 5.58$ on the last ring ($60.0 \to 65.6$), each positive in 5 of 5 seeds, and $+1.31 \pm 0.40$ without CNOTs, below half the last-ring gain. Seed-mean saturation falls to at most 0.0006 in every escape-only head, and clean training accuracy rises in every cell. The MLP probe gives $+2.54$ (default) and $+1.30$ (no CNOT). Pooled over the ten seeds, the gains are $+2.39$, $+6.19$ and $+1.10$. We had expected a default gain of about $+1.5 \pm 1$.

Induced saturation. To manipulate saturation in the other direction, we added to the ring-free circuit's loss a penalty $\lambda \,\mathrm{mean}_{j<k}\,\mathrm{relu}(0.9025 - \tanh^2 z_j)$ that pushes the first $k \in \{2, 4\}$ coordinates into saturation ($\lambda = 10$, 5 seeds). The manipulation check passes ($S = 0.40$ and 0.53), and the probe falls to 59.8 ± 22.2 and 61.5 ± 10.1, which meets the committed criterion for saturation alone accounting for most of the cost at matched S. We do not count it, because no training guard was committed for this test and the penalty damaged fit: clean training accuracy fell to 71.5% and 72.9% (96.6% without the penalty), and one $k = 2$ seed never trained (head at chance, probe 20.5%). Without that seed (an exclusion we did not commit) the $k = 2$ drop is 3.8 points at $S = 0.27$, close to the ring-family cost of 3.6 interpolated linearly in S to that value, and clean training accuracy is 86.7%. The test as run cannot separate saturation from lost fit.

Generality runs. The CZ, readout and $a_j = z_j$ heads were tuned like every new head (learning rates $\{3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$ on seeds 0–2, seeds 3–4 at the rate selected on validation); every head selected 10^{-2}, the middle of the grid. The MLP probe agrees: the $a_j = z_j$ slope is -2.0 points per unit order under it (-2.1 under the linear probe). Under $a_j = z_j$ the identity of Equation (1) holds with $\cos z_j$; its fraction "S" is not comparable to the $\pi \tanh$ heads', because nothing bounds the scale of z.

N. Effective Order Beyond Zero Rotations

Product order is defined at zero rotations. Table 23 extends it to any rotations with a nonzero readout by expanding each readout operator in Pauli strings and counting, with weight c_P^2, the inputs that each string visible on the encoded state touches; the test suite checks that the expansion reproduces the simulator's readouts and input dependences exactly, and that it equals Equation (1) at $\theta = 0$. Three observations follow. First, layouts with a single CNOT layer after the last rotations cannot change their effective order at any rotations where the readout is nonzero, because single-qubit rotations do not change which qubits a string acts on; only every-layer layouts can move, and of these brickwork moves most, from 3.5 to 2.6 at its trained rotations, below the distance-2 ring's 3.0, which restores that pair's probe ranking and shrinks its residual from the fit from $+5.1$ to $+2.3$ points. Second, at random rotations $\theta \sim \mathcal{U}(-\pi, \pi)$ the default ring's readouts depend on all eight inputs (effective order 5.4) while the ring-free ones stay at one, consistent with the gap at large initialization (Table 14). Third, random-rotation order predicts probe accuracy worse than order at $\theta = 0$ (Spearman -0.71 vs -0.91): the star and the open chain reach 3.4 and 4.2 at random rotations yet probe near order-1 circuits, because trained circuits stay close to their zero-rotation regime. Order computed at the trained rotations ranks the circuits slightly better (-0.94), but it uses the trained weights and is therefore a post hoc description, not a prediction.

Table 23. Effective order beyond zero rotations (CIFAR-10, 4,000 images, small CNN; *: second family). For rotations θ, readout k is $\langle\psi(z)|O_k|\psi(z)\rangle$ with $O_k = U(\theta)^\dagger Z_k U(\theta) = \sum_P c_P P$ over Pauli strings; strings with a Y factor vanish on the encoded state, and an X or Z factor on qubit j makes the readout depend on input j. The effective order is the c_P^2-weighted mean number of inputs a visible string touches, averaged over readouts; at $\theta = 0$ it is the product order of Equation (1). Trained: the trained rotations (5 seeds; 3 for the 1- and 2-layer rings). Random: $\theta \sim \mathcal{U}(-\pi, \pi)$ (20 draws), with the share of weight on invisible (Y) strings. Spearman correlation with probe accuracy over the 13 layouts: -0.91 at $\theta = 0$, -0.94 trained (post hoc), -0.71 random.

	Effective order				
Layout	$\theta = 0$	trained	random	Y share (random)	Probe
no CNOT	1.00	1.00	1.00	0.33	73.6
pairs	1.00	1.06	1.46	0.45	72.9
pairs, last	1.50	1.50	1.50	0.42	71.2
ring (default)	2.50	2.65	5.44	0.90	69.8
ring, 2 layers	5.25	5.12	5.39	0.80	61.2
ring, 1 layer	5.25	5.25	5.25	0.00	56.8
ring, last	5.25	5.25	5.25	0.85	60.1
star*	1.00	1.01	3.41	0.75	73.1
chain*	1.50	1.55	4.16	0.84	71.4
star-in, last*	1.88	1.88	1.88	0.40	73.0
ladder-2, last*	3.00	3.00	3.00	0.72	66.6
brickwork*	3.50	2.62	2.88	0.68	70.5
chain, last*	4.50	4.50	4.50	0.78	58.2

O. Pretrained-Backbone Controls

On the pretrained backbone the high-order heads are also unstable across runs (test accuracy spans up to 27 points over three seeds at one learning rate). In the main grid the classical head on the default circuit's products probes above the circuit (-4.0 ± 1.6 vs -7.2 ± 2.3 against linear, every seed), each at its validation-selected rate. When the circuit, the same circuit frozen at zero rotations and the products head are trained in lockstep, from copied weights on identical batches (below), they do not differ detectably at a shared learning rate (circuit − products -0.6 ± 2.0 and $+0.6 \pm 3.9$ at 3×10^{-4} and 10^{-3}, 10 seeds; 95% intervals $[-2.0, 0.8]$ and $[-2.1, 3.4]$), while at the main-grid rates the products head stays ahead ($+2.9 \pm 3.0$, 8 of 10 seeds). Learning-rate selection is thus the more likely source of the products head's advantage, without excluding a contribution of the rotations.

Table 24 separates training size from the number of optimizer updates on the pretrained ResNet-18, and Table 25 gives the lockstep comparison of the trained circuit, the circuit frozen at zero rotations and the products head discussed in Section 5.6. A step-by-step check (seed 0, head lr 10^{-3}) shows why the lockstep pairs still differ: two copies of the products head trained on the same batches stay bit-identical, while the frozen circuit and the products head, whose outputs agree to floating-point rounding (a test checks 10^{-5}), reach a backbone parameter distance of 2×10^{-5} after one update, 0.14 after 21 and 3.7 after 125 (one epoch), with the backbone's parameter norm at 85.

Table 24. Training budget on the pretrained ResNet-18 (CIFAR-10; seeds 0–2; probe accuracy, %). The protocol trains 30 epochs at every size, so smaller training sets get fewer optimizer updates. Matched rows train 120 epochs at 1,000 images and 12 at 10,000, approximately the update count of 4,000 images × 30 epochs (see the updates column), with the head learning rates selected at 30 epochs for each size or re-selected on validation over a half-decade step either way (re-tuned). Differences are paired over seeds, against the linear head.

Images	Epochs	Updates	linear	default − linear	no CNOT − linear
1,000	30	960	82.9 ± 0.9	-22.3 ± 2.9	-1.6 ± 1.2
1,000	120 (matched)	3,840	81.7 ± 0.9	-6.6 ± 1.7	$+0.2 \pm 1.3$
1,000	120 (matched, re-tuned)	3,840	83.3 ± 0.6	-8.3 ± 2.1	-1.4 ± 1.2
4,000	30	3,750	89.5 ± 0.3	-5.7 ± 0.7	0.0 ± 0.2
10,000	30	9,390	92.6 ± 0.4	-1.9 ± 0.4	-0.1 ± 0.4
10,000	12 (matched)	3,756	91.4 ± 0.3	-7.2 ± 2.1	$+0.5 \pm 0.2$
10,000	12 (matched, re-tuned)	3,756	92.6 ± 0.1	-7.7 ± 1.9	-0.7 ± 0.2

Table 25. Lockstep control on the pretrained ResNet-18 (CIFAR-10, 4,000 images). The trained default circuit, the same circuit with its rotations frozen at exactly zero (forward map and input gradients equal to the products head's), and the classical head on the default circuit's zero-rotation products are trained in one process from copied initial weights on identical batches; paired probe differences over seeds 0–9 at each head lr, with a 95% paired t interval. The last row pairs the heads at the rates validation selected for them in the main grid. Mean probe accuracy (trained / frozen / products), 3×10^{-4}: 82.0 / 82.7 / 82.6; 10^{-3}: 85.6 / 83.9 / 84.9.

Paired difference	mean ± SD	95% interval	negative
trained − products, both at 3×10^{-4}	-0.6 ± 2.0	[-2.0, +0.8]	5/10
frozen − products, both at 3×10^{-4}	0.0 ± 0.6	[-0.4, +0.5]	4/10
trained − products, both at 10^{-3}	$+0.6 \pm 3.9$	[-2.1, +3.4]	5/10
frozen − products, both at 10^{-3}	-1.0 ± 2.6	[-2.9, +0.8]	6/10
trained at 3×10^{-4} − products at 10^{-3} (main-grid rates)	-2.9 ± 3.0	[-5.1, -0.7]	8/10

P. SVHN, Data Re-uploading and a Second Pretrained Backbone

Three further tests had predictions and pass criteria committed before their additional runs (Table 26); as committed, the second SVHN test reuses the first test's runs wherever a head's selected rate is unchanged (20 of its 60 scored runs). Each head's learning rate was selected on the mean validation accuracy of seeds 0–2, with one half-decade step added at any grid edge; every selected rate is interior. The primary measure is the common linear probe; the MLP probe changes no verdict.

SVHN. A first test (5 seeds) left the no-CNOT and pairs heads at the lower edge of their grids, so its order comparisons were inconclusive; its gaps were driven by runs that collapsed to the majority class (head validation accuracy below 25%), in four of five last-ring seeds. A second, separately committed test extended the grids downward and used 10 seeds. At the newly selected rates no run collapses, and the order holds seed by seed: default − no CNOT -12.8 ± 3.8 and last ring − default -18.1 ± 9.5, every seed negative, while the pairs circuit, which entangles at order 1, probes 13.6 ± 3.4 points above the default (every seed) and 0.9 ± 1.8 above the ring-free head. The linear head is 14.9 ± 4.3 above the default. These pass; the prediction that collapses would persist and order the heads (no CNOT $\leq$ default $\leq$ last ring, with at least three more for the last ring than for no CNOT) fails, because there are none. The classical products head is 1.6 ± 9.2 below the default circuit, within the committed two points.

Data re-uploading. With the encoding applied before each of the four variational layers the zero-rotation identity no longer holds, and we predicted only that the single-encoding ordering would carry over. It does: default − no CNOT -10.0 ± 3.2 and last ring − default -9.7 ± 7.1 (every seed negative), and the four layouts rank exactly by their single-encoding orders (Spearman -1). Re-uploading lowers every layout against its single-encoding counterpart (unpaired; no CNOT -5.6, pairs after the last layer -4.9, default -11.8, last ring -11.8), the two ring layouts about twice as much, and saturation again rises with order.

MobileNetV3-Small. On a second ImageNet-pretrained backbone, with the ResNet-18 recipe, the direction repeats but the cost is much smaller. The pairs circuit is 1.5 ± 0.6 above the default and the linear head 1.6 ± 0.8 above it, every seed, and pairs is 0.6 ± 0.6 above the ring-free head (4 of 5 seeds; pass). The order test fails on one part: default − no CNOT is -0.9 ± 1.0, negative in every seed but short of the committed -1.0, while last ring − default is -5.5 ± 5.8 (every seed). The default circuit saturates little on this backbone ($S = 0.13$, against 0.30–0.49 for ring circuits on the small CNN).

What collapse is. A fourth committed test asked whether order causes the SVHN collapses (the collapse counts of the runs above were seen before it was written, and it predicts nothing about them). Differentiating Equation (1) at zero rotations, with independent coordinates, gives each input's readout sensitivity a factor $\prod_{i \in S_k \setminus \{j\}} c_i$ with $c_i = \mathbb{E}[\cos^2 \varphi_i] \leq 1$: higher order adds derivative paths but attenuates each one. The committed measurement on the untrained network fails both predictions (seed-mean ratio of measured to predicted sensitivity in $[0.75, 1.33]$ for every layout; no CNOT > default > last ring in every seed). That measurement, however, ran BatchNorm in evaluation mode, where the untouched running statistics leave the bottleneck near zero (mean $|\tanh z| = 0.03$), not with the batch statistics of the first training steps. Repeated with batch statistics (batches of 32; not committed in advance), the bottleneck is wider (mean $|\tanh z| = 0.23$, $c = 0.60$) and measured over predicted sensitivity lies between 0.89 and 1.13 for every layout and seed. The last ring then has the lowest summed squared readout gradient in all 5 seeds (16.7, against 23.6 for no CNOT and 24.8 for the default ring), but the default ring is not below no CNOT (above it in 4 of 5 seeds), so the committed order still fails. Over all 141 SVHN

circuit and products runs (41 collapsed), collapse goes with a dying backbone gradient: the ratio of the mean backbone gradient norm over epochs 2–10 to its epoch-1 value separates collapsed from other runs with AUC 0.94 (0.91–1.00 within each layout), with medians 0.12 and 0.76, though the two ranges overlap. This is an association over the epochs in which collapse happens, not a test of which comes first. At an untried rate (5×10^{-3}, 6 seeds), collapses number 2, 2, 2 and 5 for no CNOT, pairs, default and last ring. This meets the committed ordering, but only the last ring stands out, and the default ring's non-collapsed runs reach 26–40% validation accuracy, against 61–86% for no CNOT.

Q. Exploratory Sweep with One Shared Recipe

Before the protocol of Section 3, we ran every head with one shared recipe: head learning rate 10^{-3} (backbone 5×10^{-4}), 25 epochs on MNIST and 80 on CIFAR-10, three data re-uploading schedules for the VQC, and checkpoints chosen by best test-split accuracy. Because the test split took part in checkpoint selection, these numbers are exploratory; mean final-epoch accuracy is within 1.35 points of best-epoch accuracy in all 30 configurations and no ordering of heads changes (`scripts/make_tables.py` on the released per-epoch logs in `results/raw`). Tables 27 and 28 give the results. Table 28 is the last of three completed 80-epoch CIFAR-10 sweeps, all released. One of the other two used weight decay 10^{-3}; the other has identical recorded settings but, judging from its much larger train–test gaps, predates the random rotation and colour jitter in the augmentation (augmentation was not then recorded in run metadata, so this is an inference). We report the last because it matches the final pipeline. All three give the same ordering of heads at every size, and the weaker-augmentation sweep favours the VQC head more (62.9% against 58.9% at 4,000 images). Re-uploading lowered accuracy in every setting under this recipe (NONE > LIGHT > STRONG); it was not part of the initial tuned protocol (a separately tuned re-uploading test is in Section P).

R. Frozen-Embedding Transfer

To separate a head from the backbone it induces, we also trained fresh heads on fixed embeddings extracted from exploratory-sweep backbones (linear- or VQC-trained; MNIST at 1,000 images, CIFAR-10 at 4,000; 3 seeds; 498 head fits with learning rate and epoch selected on validation). Table 29 gives the results. On VQC-trained embeddings, a refitted linear head beats a refitted VQC by 1.7 ± 0.8 (MNIST) and 1.6 ± 0.5 points (CIFAR-10), positive for every seed. On linear-trained embeddings, most of whose values lie in the flat region $|\tanh z| > 0.9$ of the encoding (56% and 66%), the VQC collapses, trailing the linear head by 20.9 ± 4.4 and 41.8 ± 2.9 points, and a classical Fourier head $[\cos a_j, \sin a_j] \to$ Linear with the same encoding collapses almost as far on CIFAR-10. Fixed standardization of the embedding recovers 12.6 ± 6.2 and 30.4 ± 2.9 of those points for the VQC, and a learned per-dimension affine 15.0 ± 7.6 and 35.9 ± 0.9; tripling the training budget recovers 0.5 ± 0.2 on CIFAR-10. The Fourier head's recovery under standardization was not tested. The source checkpoints were selected on the test split, so this experiment is exploratory; it motivated the probe measurements of Section 5, which use validation-selected backbones only.

S. Learning-Rate Sensitivity

Trigonometric heads, quantum or classical, tolerate a narrow range of head learning rates under the coupled recipe (backbone at half the head rate). On CIFAR-10/4,000 all of them select 10^{-2}, and at 3×10^{-2} test accuracy falls to 63.4 for the default VQC, 50.0 for the ring-free VQC, and 29.3 and 28.9 for the classical cos and cos+sin heads, while the linear head selects 3×10^{-2}. With the head learning rate fixed, the default VQC also collapses at backbone learning rates of 3×10^{-2} and 6×10^{-2}, where its bottleneck saturates the encoding almost completely, whereas the linear head still trains (Table 10). Test accuracy for every grid point is in the released results.

T. Sign Symmetry

At zero rotations every readout is an even function of each z_j, because cosine is even. We measured how far trained heads are from this symmetry on CIFAR-10/4,000 test embeddings: the median, over samples, of $\|q(-\mathbf{z}) - q(\mathbf{z})\| / \|q(\mathbf{z}) - q(\mathbf{z}')\|$, with q the readouts and $\mathbf{z}'$ a random other sample (0: blind to the sign; about 1: flipping the sign moves the readout as far as changing the image). The frozen circuit stays nearly sign-blind (0.09–0.12), and its accuracy on $-\mathbf{z}$ is within 1.4 points of that on $\mathbf{z}$. Trained heads are not: 0.58–0.73 for the default VQC (accuracy on $-\mathbf{z}$: 25–34%), 0.91–1.02 for the ring-free VQC (6–17%), and 0.85–0.97 for the monotone and symmetric encodings of Table 10. Sign-blindness at initialization therefore does not predict the backbone cost, consistent with the null result of the monotone encoding.

U. Training and Reproducibility Details

Training. AdamW with weight decay 10^{-4}, cosine annealing to zero, batch size 32, gradient clipping at norm 1.0 (MNIST) or 2.0 (CIFAR-10); CIFAR-10 augmentation is random crop, horizontal flip, small rotation and color jitter. Head learning-rate grids: MNIST $\{3 \times 10^{-4}, \ldots, 3 \times 10^{-2}\}$ and CIFAR-10 $\{10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$ in half-decade steps, each extended by one step whenever the selected rate was at the top; circuit variants $\{3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$; classical product heads $\{10^{-3}, \ldots, 3 \times 10^{-2}\}$ on CIFAR-10 circuit supports and $\{3 \times 10^{-3}, \ldots, 3 \times 10^{-2}\}$ on designed supports and on MNIST (extended to 10^{-1} for the MNIST last-ring products). All selected rates of circuit and classical product heads are interior to their grids, except in the re-tuned matched-budget search on the pretrained backbone (one half-decade step either way), where the 10,000-image default head and the linear head at both sizes select a boundary rate; that search is local. The probe uses AdamW (weight decay 10^{-4}), batch size 128, 60 epochs with a cosine schedule, and probe learning rates $\{3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}, 10^{-1}, 3 \times 10^{-1}, 1\}$, all selected on validation.

Reproducibility. The repository contains per-epoch metrics and metadata for every training run, the probe, frozen-head and trained-circuit measurements, and the scripts that compute every number and generate most tables and figures from them (`scripts/summarize_v2.py`, `scripts/zero_angle_check.py`, `paper/figures/scripts/`). The PyTorch statevector simulator is tested against PennyLane (absolute tolerance 10^{-6} in outputs and input gradients, 10^{-5} in rotation gradients; relative 10^{-5}); the test suite also checks the zero-rotation identity, the ring-free form, the entangling example of Section A, the equality of the zero-rotation circuit and the products head, and the effective-order expansion of Section N against the simulator. The predicted orderings for the MNIST replication and the predictions and pass criteria for the second family of layouts were written in the project's log before those runs; the log entries are copied verbatim into the repository, whose first timestamp for them postdates the runs, so the earlier dates are our own report. The predictions for the two encoding arms (Sections H and I) were committed before their runs. Seeds control data subsets, initialization and data order. Experiments ran on one Apple M3 Max. The reviews and the claim and citation audits of this paper were run with the ARIS workflow (Yang et al., 2026).

Table 26. Pre-registered extensions (common linear probe, test accuracy in %, mean ± SD over n seeds at each head's validation-selected learning rate, every selected rate interior to its grid; Δ: paired difference to the head named in the column). SVHN: 4,000 training images, small CNN, CIFAR-10 recipe without flips, 10 seeds. Re-uploading: CIFAR-10/4,000, small CNN, encoding applied before each of the 4 layers, 5 seeds; orders are those of the single-encoding layouts. MobileNetV3-Small: ImageNet-pretrained, CIFAR-10/4,000, recipe of Table 4, 5 seeds. S: fraction of bottleneck coordinates with $|\tanh z| > 0.9$ (not given for the linear head, which has no encoding).

Head	Order	Probe	S	Δ
SVHN ($n = 10$); Δ to no CNOT				
linear	—	83.4 ± 5.1	—	$+2.1 \pm 5.6$
VQC, no CNOT	1	81.2 ± 2.0	0.02	—
VQC, pairs	1	82.1 ± 1.3	0.02	$+0.9 \pm 1.8$
cos-prod. (default supports)	2.5	66.9 ± 7.9	0.57	-14.4 ± 7.3
VQC, ring (default)	2.5	68.5 ± 2.8	0.25	-12.8 ± 3.8
VQC, ring, last	5.25	50.4 ± 9.1	0.55	-30.9 ± 9.7
Re-uploading, CIFAR-10 ($n = 5$); Δ to no CNOT				
VQC, no CNOT	1	68.0 ± 2.1	0.18	—
VQC, pairs, last	1.5	66.2 ± 2.0	0.22	-1.8 ± 1.2
VQC, ring (default)	2.5	58.0 ± 4.0	0.53	-10.0 ± 3.2
VQC, ring, last	5.25	48.3 ± 4.6	0.61	-19.7 ± 5.8
MobileNetV3-Small, CIFAR-10 ($n = 5$); Δ to linear				
linear	—	86.1 ± 0.6	—	—
VQC, no CNOT	1	85.4 ± 0.6	0.18	-0.7 ± 0.7
VQC, pairs	1	86.0 ± 0.3	0.15	-0.0 ± 0.3
VQC, ring (default)	2.5	84.5 ± 0.7	0.13	-1.6 ± 0.8
VQC, ring, last	5.25	79.0 ± 5.9	0.33	-7.1 ± 5.5

Table 27. Exploratory sweep, MNIST: best test-split accuracy (%, mean ± SD over 3 seeds) with one shared recipe. Bold marks the better of MLP-8 and the VQC (NONE).

			VQC (re-uploading schedule)		
N_{train}	Linear	MLP-8	NONE	LIGHT	STRONG
250	61.5 ± 1.7	36.7 ± 5.4	$\mathbf{54.5 \pm 3.7}$	48.3 ± 10.7	38.3 ± 7.4
500	79.3 ± 2.9	53.2 ± 6.9	$\mathbf{71.2 \pm 1.9}$	57.8 ± 2.5	45.6 ± 6.7
1,000	89.6 ± 0.7	76.4 ± 3.4	$\mathbf{83.3 \pm 0.3}$	68.4 ± 3.5	54.6 ± 14.0

Table 28. Exploratory sweep, CIFAR-10: best test-split accuracy (%, mean ± SD over 3 seeds) with one shared recipe. Bold marks the better of MLP-8 and the VQC (NONE).

			VQC (re-uploading schedule)		
N_{train}	Linear	MLP-8	NONE	LIGHT	STRONG
1,000	57.5 ± 0.8	$\mathbf{51.2 \pm 0.8}$	37.1 ± 2.4	34.2 ± 0.7	26.6 ± 1.9
2,000	64.6 ± 0.7	$\mathbf{60.6 \pm 0.9}$	46.8 ± 5.2	42.7 ± 3.9	28.7 ± 1.9
4,000	71.2 ± 0.5	$\mathbf{67.2 \pm 0.4}$	58.9 ± 4.6	52.0 ± 7.8	35.2 ± 5.5

Table 29. Frozen-embedding transfer: test accuracy (%, mean ± SD over 3 seeds) of heads trained on identical frozen embeddings. Columns give the head the backbone was trained with and the fraction of test embedding values with $|\tanh z| > 0.9$. †Selected learning rate is the largest tried for that head.

	MNIST (1,000 images)		CIFAR-10 (4,000 images)	
Head	linear src. (56%)	VQC src. (5%)	linear src. (66%)	VQC src. (30%)
Linear	89.8 ± 0.2	88.9 ± 1.6	70.9 ± 0.6	$62.5 \pm 4.4^{\dagger}$
MLP-8	88.3 ± 0.9	88.5 ± 1.1	70.2 ± 0.8	62.3 ± 4.0
Fourier	78.2 ± 1.1	87.1 ± 2.0	33.4 ± 1.0	61.8 ± 4.5
VQC	69.0 ± 4.4	87.2 ± 1.0	29.1 ± 3.0	61.0 ± 4.1
VQC+std	81.6 ± 5.6	77.6 ± 6.6	59.5 ± 0.4	53.7 ± 5.6
VQC+std+affine	84.0 ± 3.3	82.7 ± 1.6	65.0 ± 3.1	61.6 ± 4.4
VQC, 3× epochs	–	–	$29.6 \pm 3.0^{\dagger}$	$61.0 \pm 4.5^{\dagger}$